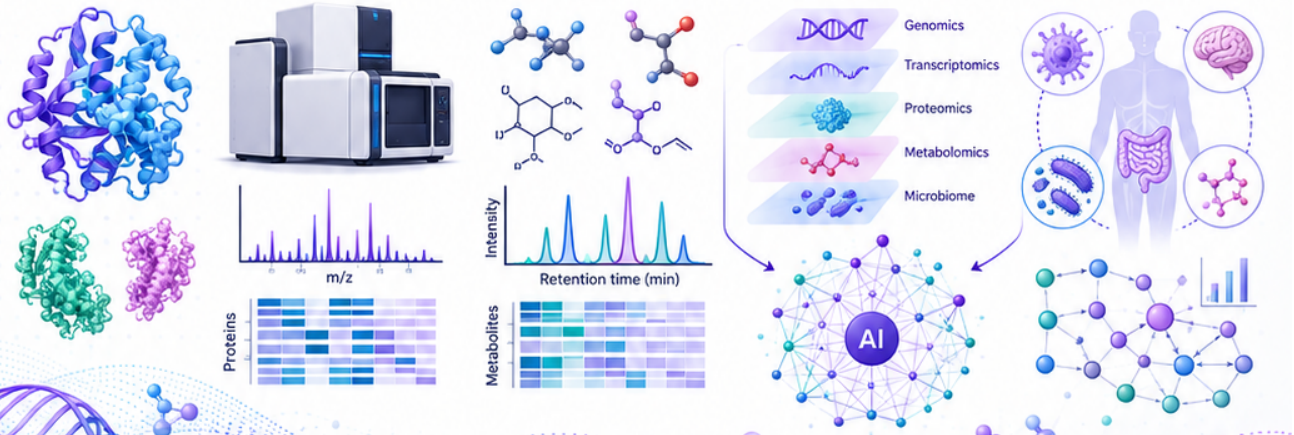
BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

WEEK 7 READING MATERIAL

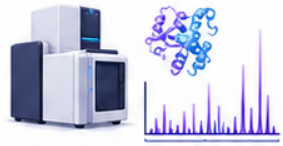
Functional Omics: Proteomics & Metabolomics

From measurement to mechanism, hypothesis, and multi-omics integration

PROTEINS → PROTEOMICS → METABOLOMICS → MULTI-OMICS INTEGRATION → BIOLOGICAL INTERPRETATION

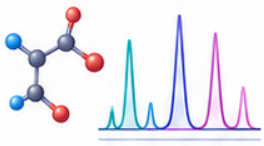


PROTEOMICS



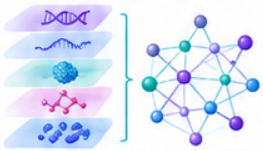
Measure protein abundance, complexes, and post-translational modifications.

METABOLOMICS



Profile small molecules that reflect pathway output and host–microbe interactions.

INTEGRATION



Link proteins and metabolites with genomics, transcriptomics, and microbiome data.

INTERPRETATION

Turn molecular measurements into testable biological hypotheses.

Fall 2026 | Two 40-Minute Undergraduate Sessions | Dr. Liwei Xie

How to use this handout

This Week 7 handout is designed for third-year undergraduates who have already completed the genomics, transcriptomics, and microbiome units. The central teaching goal is not only to introduce proteomics and metabolomics as new data types, but to help students understand why these two functional layers are indispensable for building biological explanations, refining scientific questions, and deciding what experiment should come next.

Read the handout in the following order. First, understand how proteomics and metabolomics extend the logic of the previous weeks. Second, learn the difference between targeted and untargeted strategies, because nearly every experimental design decision depends on that distinction. Third, study the quality-control logic before memorizing any plot or software name. Fourth, pay special attention to the integration sections, because the real scientific value of functional omics emerges when protein and metabolite data are interpreted together and then linked back to DNA, RNA, and microbes. Finally, use the AI-agent section as a workflow scaffold, not as a replacement for scientific thinking.

- Core questions of Week 7: What is the cell actually doing? Which proteins execute the response? Which metabolites report pathway output or host-microbe communication?
- Core methodological decision: discovery-oriented untargeted profiling versus hypothesis-driven targeted measurement.
- Core analytical rule: quality control comes before differential analysis, and differential analysis comes before mechanistic interpretation.
- Core integrative rule: a cross-omics correlation is only the start of a hypothesis, not the end of a story.
- Core AI rule: let agents automate operations, but keep human control over biological design, statistical validity, evidence grading, and claim boundaries.

Learning outcomes

1. Explain how proteomics and metabolomics extend the logic of genomics, transcriptomics, and microbiome profiling.
2. Distinguish targeted and untargeted strategies in both proteomics and metabolomics, and justify when each is appropriate.
3. Describe key protein-centered combinatorial assays, including IP/Co-IP-MS, affinity purification-MS, proximity labeling proteomics, RIP/CLIP-style RNA-binding assays, and ncRNA-integrated workflows.
4. Interpret raw-output quality control, data preprocessing, statistical testing, pathway analysis, and multi-omics integration in a hypothesis-driven manner.
5. Use AI agents such as ChatGPT-, Gemini-, or Claude-based workflows to assist coding and interpretation while preserving scientific rigor and reproducibility.

0 | Why Week 7 matters after genomics, transcriptomics, and microbiome analysis

Genomics taught us how to measure relatively stable information: DNA sequence variation, regulatory elements, chromatin state, and the potential architecture of the system. Transcriptomics moved one layer closer to function by quantifying RNA abundance, isoforms, and noncoding RNAs, thereby revealing which programs are being transcribed under a given condition. Week 6 on microbiome analysis added a parallel biological system: microbial taxa, ecological structure, and community gene content. Yet all three earlier layers still leave a critical question unresolved: what is the integrated functional state of the host-microbe system?

Proteomics and metabolomics answer this question from complementary directions. Proteomics measures the molecules that execute work: enzymes, receptors, transporters, transcription factors, signaling complexes, and post-translationally modified protein states. Metabolomics measures the small molecules produced, consumed, secreted, or transformed by host cells and microbes, including amino acids, lipids, bile acids, short-chain fatty acids, central-carbon intermediates, and drug-related compounds. In short, proteomics helps answer "which machinery is operating," whereas metabolomics helps answer "what chemical state has emerged."

These layers are especially important for the themes emphasized throughout this course. A genomic variant may alter a regulatory region, but only proteomics can show whether the encoded pathway is actually activated. An RNA-seq signature may suggest inflammatory activation, but phosphoproteomics can identify the receptor pathway and downstream kinase events. Microbiome sequencing may reveal bile-acid metabolism genes or tryptophan-pathway capacity, but metabolomics is required to confirm whether those microbial functions are reflected in measurable host-associated small molecules. Therefore, Week 7 is the functional extension of the earlier weeks, not a separate topic.

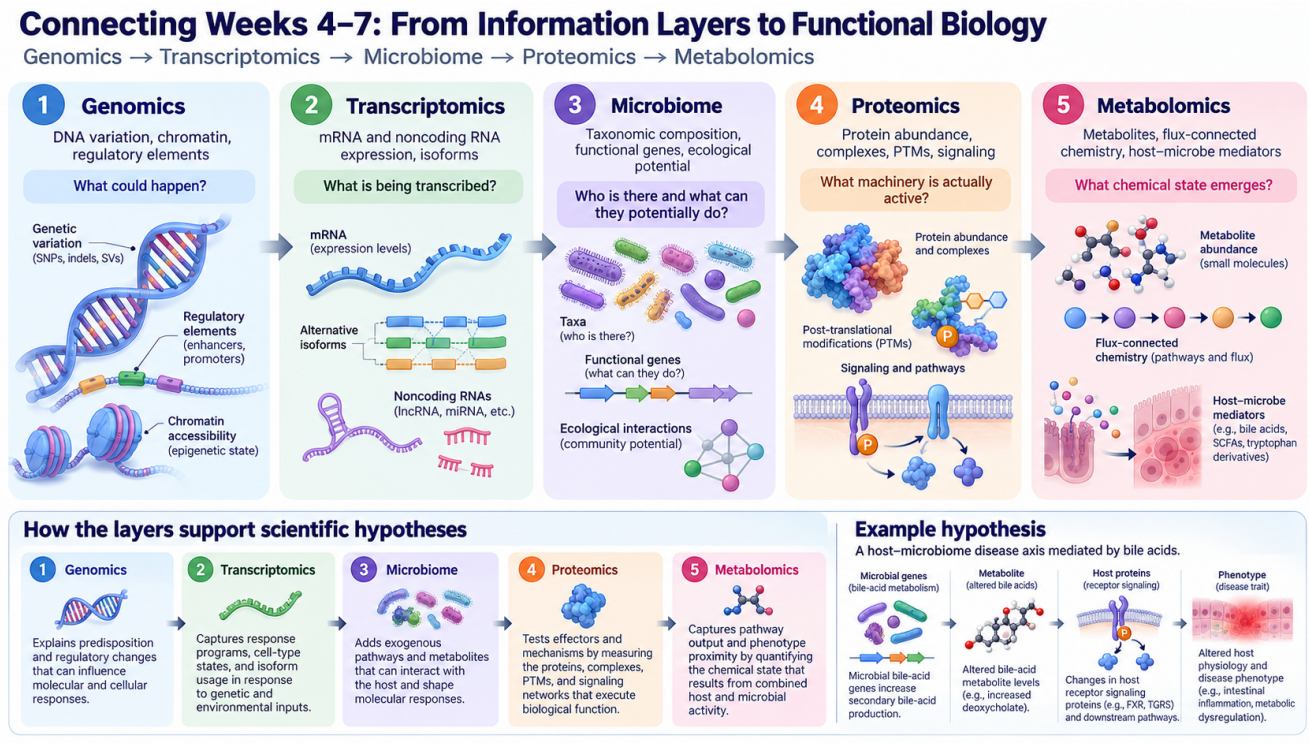


Figure 1. Conceptual bridge across Weeks 4-7. Genomics, transcriptomics, microbiome, proteomics, and metabolomics are not separate silos; they are linked information layers that progressively approach mechanism and phenotype.

0.1 From regulatory DNA to functional protein

A noncoding variant in an enhancer can alter chromatin accessibility or transcription-factor binding and thereby change an RNA isoform. In the Week 4-5 toolkit, whole-genome sequencing locates the variant, ATAC-seq characterizes accessible chromatin, ChIP/CUT&Tag detects DNA-associated factors or histone marks, and RNA-seq measures transcript abundance and splicing. Week 7 asks whether the variant alters the abundance, localization, modification or interaction partners of the encoded protein. The expression of a gene may be unchanged while a protein becomes phosphorylated, moves to a membrane compartment or joins a signaling complex. A follow-up could therefore combine total proteomics, site-resolved phosphoproteomics and a receptor-focused Co-IP-MS experiment.

When combining genome and proteome measurements, preserve variant identifiers and reference assemblies. Variant-derived peptides require a sample-specific sequence database and adequate peptide-spectrum evidence; an amino-acid substitution predicted from DNA is not automatically detected at the protein level. A proteogenomic discovery provides direct evidence only for the peptides actually supported by MS spectra, not for every predicted open reading frame.

0.2 From transcript and noncoding RNA to protein output

Total RNA-seq can detect transcript abundance and alternative splicing, whereas small-RNA-seq characterizes small regulatory RNAs. Polysome profiling and ribosome profiling probe translation more directly, and proteomics then measures protein accumulation. These assays answer different questions: transcriptional regulation, translation engagement and steady-state protein abundance are separable. Proteomic discordance can arise from translational repression, differences in protein half-life, secretion, selective degradation or cell-type composition. Never cite a universal mRNA-protein correlation value without defining tissue, conditions, normalization and the unit of comparison.

For a miRNA-centered question, measure candidate miRNA and target transcript levels, perform a validated perturbation such as mimic/inhibitor or genetic modulation, quantify the relevant protein and use a 3′UTR reporter with seed-site controls. For an lncRNA-centered question, distinguish an RNA-protein interaction from changes in a protein complex: RNA pull-down-MS or ChIRP-MS asks what binds the RNA; RIP/eCLIP asks which RNAs bind a selected protein; conventional Co-IP-MS asks which proteins associate with a bait. The combined study must demonstrate specificity and ideally an effect on translation, localization, protein stability or a downstream phenotype.

0.3 From microbiome potential to metabolite-mediated host response

Amplicon data (16S/18S/ITS) provide marker-dependent taxonomic information rather than direct functional activity. Shotgun metagenomics measures microbial genes and community potential; metatranscriptomics reports microbial RNA expression; metaproteomics provides evidence of expressed microbial proteins; targeted and untargeted metabolomics quantify or discover chemical products. Host proteomics and phosphoproteomics then measure candidate host receptor pathways, transporters and effector mechanisms. A gene for bile-acid transformation in a stool metagenome does not prove that a particular serum bile acid is microbially derived: substrate supply, expression, enzyme activity, host processing and transport all matter.

Consider an initially descriptive chain: microbial bile-acid transformation capacity → chemically authenticated secondary bile acid → host receptor-signaling response → lipid-metabolic phenotype. Each arrow is a separate inference requiring its own measurement or intervention. Compare paired fecal and circulating metabolomes, use target-specific host readouts, account for diet and medication, and consider time-resolved sampling. A microbial genetic perturbation plus metabolite rescue and host-receptor blockade is much more informative about directionality than cross-sectional correlations alone.

0.4 Translating a biological question into an assay design

Research question	Previous-week assay	Week 7 functional readout	Next experiment
Does a noncoding variant alter signaling?	WGS + ATAC/CUT&Tag + RNA-seq	Total and phosphoproteomics	Allele editing; receptor blockade
Does a lncRNA scaffold a protein complex?	RNA-seq / ncRNA perturbation	RNA pull-down-MS; RIP/eCLIP; Co-IP-MS	RNA truncation + binding-site rescue
Does a gut bacterium produce a host-active compound?	Metagenomics + microbial RNA	Targeted metabolomics + host proteomics	Defined culture, isotope tracing, gnotobiotic testing
Does an engineered enzyme improve product formation?	Genome edit + transcriptomics	Enzyme proteomics + targeted product / flux	DBTL iteration: promoter or enzyme redesign
Can a candidate biomarker be measured reproducibly?	Variant / RNA candidate discovery	PRM or targeted panel + calibrated LC-MS	Blinded external cohort and orthogonal validation

1 | Targeted and untargeted strategies in functional omics

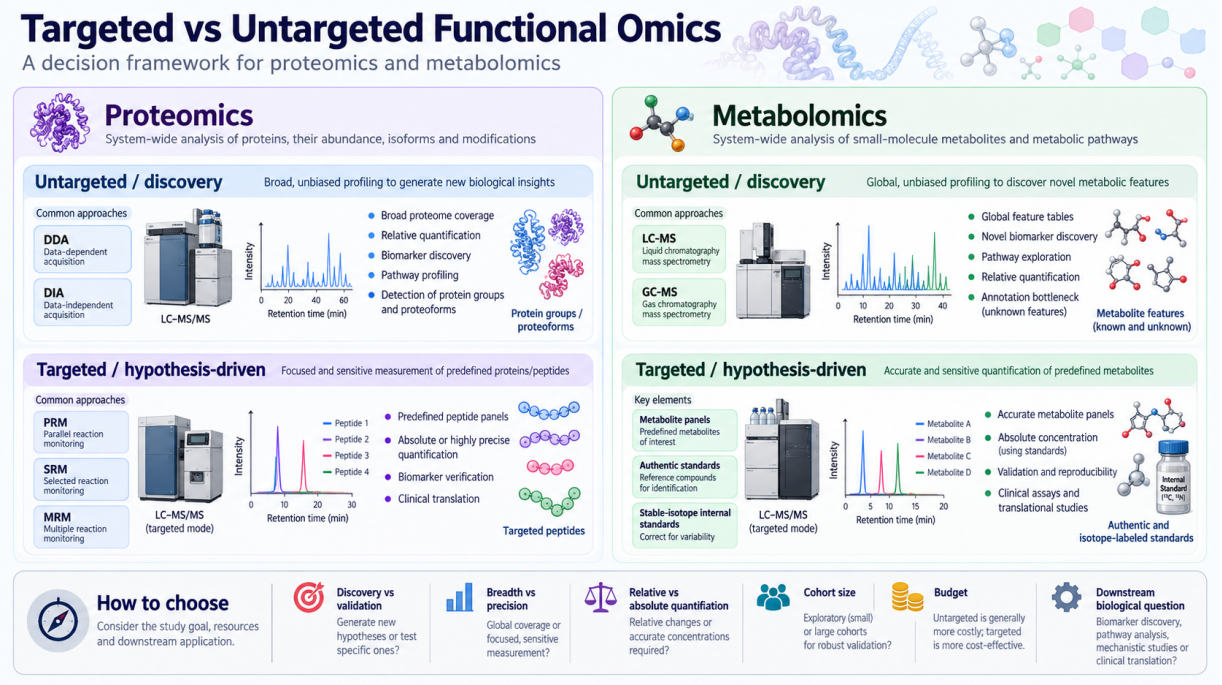


Figure 2. Targeted versus untargeted functional omics in proteomics and metabolomics. Discovery designs maximize breadth; targeted designs maximize sensitivity, precision, and translational robustness.

Both proteomics and metabolomics can be performed in untargeted or targeted modes, and this is the first major design decision students should learn to make. Untargeted approaches are discovery-oriented. They attempt to measure as many proteins, peptides, or metabolic features as possible, usually producing relative rather than absolute quantification. Their main strengths are breadth, hypothesis generation, and the ability to discover unexpected biology. Their main weaknesses are more missingness, more complex annotation, and a greater burden of statistical correction and validation.

Targeted approaches are hypothesis-driven. They focus on a predefined panel of analytes and typically use a more restricted but better-controlled measurement strategy. Their strengths are higher sensitivity, stronger reproducibility, better comparability across runs or cohorts, and easier clinical translation. Their weakness is that they cannot discover what they were not designed to measure.

- Untargeted proteomics: commonly implemented using DDA or DIA workflows for broad protein coverage, comparative biology, biomarker discovery, and pathway exploration.
- Targeted proteomics: commonly implemented using PRM, SRM, or MRM to quantify a predefined set of peptides or proteins, often in validation cohorts or mechanism-focused studies.
- Untargeted metabolomics: commonly implemented using LC-MS or GC-MS global profiling, generating thousands of features that require quality filtering and careful annotation.
- Targeted metabolomics: commonly implemented using authentic standards and stable-isotope internal standards to measure a predefined panel such as bile acids, SCFAs, amino acids, acylcarnitines, or drug metabolites.

In practice, discovery and validation are often linked in a two-stage strategy. Students should recognize the standard progression: untargeted discovery in a smaller or mechanistic cohort, followed by targeted verification in an independent cohort or targeted perturbation experiment. This logic applies equally well to protein biomarkers, phosphosite panels, microbial metabolite mediators, and host-microbe communication pathways.

1.1 Acquisition, labeling, and analytical platform decisions

Approach	What it measures	Practical decision	Important limitation
DDA discovery	Selected precursors followed by MS2	Flexible library generation; discovery / specialized PTMs	Sampling can vary between runs
DIA discovery	Repeated systematic isolation windows	Consistent quantitative cohorts; library-based or library-free	Chimeric spectra require careful interference control
LFQ label-free	LC-MS peptide intensities by run	Flexible sample number, relatively low labeling cost	Run-order and batch monitoring essential
TMT multiplexing	Isobarically tagged sample multiplexes	High-throughput comparative designs	Ratio compression, channel balancing, inter-batch bridges
SILAC	Isotopic amino-acid incorporation	Cultured-cell mechanistic perturbations	Not a routine direct labeling approach for patient biospecimens
PRM / SRM / MRM	Prespecified peptide precursors/fragments	Independent cohort / targeted pathway verification	Limited scope; peptide choice and interference matter
Immunoassay / affinity panel	Antibody/aptamer-dependent protein epitopes	Large biofluid cohorts; clinical screening	Binding specificity and proteoform cross-reactivity

DDA and DIA describe precursor-selection and fragmentation strategies; LFQ, TMT and SILAC describe how samples are labeled or compared. Do not write “DIA is targeted metabolomics” or “TMT is synonymous with DDA.” A DIA run can be quantified without an empirical spectral library when a suitable prediction or search workflow is available. A TMT assay may use MS2 and/or MS3-based methods; study-specific validation is required to assess interference and ratio compression. Precision, coverage and missing-data rates depend on specimen, instrument, acquisition schedule, identification model and analysis settings: do not memorize fixed proteins-per-run or coefficient-of-variation values as universal performance guarantees.

1.2 Targeted protein validation: selecting peptides and standards

Targeted proteomics begins with a defined biological question and a small panel of candidate proteins. Choose proteotypic peptides with unique sequences, reliable digestion, acceptable ionization, minimal variable modification, and little evidence of interfering ions. Synthetic stable-isotope labeled peptides provide internal standards to normalize signal and support calibrated quantification; full-length protein standards can also help characterize processing recovery when justified. SRM/MRM instruments monitor predefined precursor-to-fragment transitions, whereas PRM collects a broader high-resolution fragment-ion spectrum for each selected precursor. In both cases, verify coelution of transitions, relative fragment ratios, retention time, calibration linearity, blanks, lower-limit detection/quantification and intra/inter-batch precision.

A reliable validation study pre-specifies the peptide panel, training/validation partition, sample size and exclusion criteria. If a candidate biomarker is selected on the discovery cohort, selecting a new threshold on the validation cohort destroys its independent status. When an assay measures concentration, report the unit and reference material, and distinguish concentration from arbitrary normalized intensity. Targeted proteomics is not limited to biomarker work: it also validates synthetic-enzyme expression, tracks phosphorylation readouts and quantifies pathway bottlenecks in biomanufacturing.

2 | Proteomics: from abundance measurement to mechanism

A proteome is the complete set of proteins present in a specified system at a specified time. Unlike DNA, the proteome is dynamic. Unlike RNA, the proteome includes direct executors of function and can be modified after translation through phosphorylation, acetylation, ubiquitination, glycosylation, cleavage, oxidation, and many other post-translational modifications. This is why the correlation between transcript and protein abundance is informative but incomplete: transcription matters, yet translation efficiency, protein turnover, secretion, degradation, and PTM-dependent activity all contribute to the final phenotype.

In bottom-up proteomics, proteins are extracted, reduced, alkylated, enzymatically digested into peptides, separated by liquid chromatography, ionized, and analyzed by tandem mass spectrometry. The resulting spectra are matched to peptide sequences by search engines and then summarized to peptide- or protein-level quantitative matrices. DDA selects precursor ions stochastically, whereas DIA fragments all ions in a structured mass window scheme and usually provides stronger reproducibility across large cohorts. Label-free quantification is often used for broad comparison; TMT multiplexing improves throughput and reduces run-to-run variation; PRM/SRM/MRM are used when a small peptide panel must be measured precisely.

2.1 Proteome complexity, protein groups, and proteoforms

The proteome changes with cell type, time, condition, compartment and subcellular localization. A protein may occur as multiple proteoforms due to sequence variants, splice products, post-translational modifications and proteolytic processing. A protein group reported by a bottom-up search engine is not the same as a uniquely resolved proteoform. Peptide-level evidence must be inspected to distinguish paralogs and to localize a phosphorylation or other modified residue. A target-specific immunoassay, an affinity proteomics panel and untargeted LC-MS/MS do not necessarily quantify the same molecular form or recognize the same epitope.

Total-proteome studies describe relative abundance across many proteins, but the low abundance of receptors, transcription factors and some secreted mediators can require enrichment or targeted assays. Extracellular vesicles, the secretome, membrane proteome, nuclear proteome and organellar fractions each demand controls for purity, recovery and contamination. If the question concerns kinase activation, total kinase abundance and substrate phosphorylation must be interpreted separately.

2.2 Specimen-specific preparation and bottom-up mass spectrometry

Begin with a written sampling and randomization plan: biological replicates must be independent, acquisition batches must be balanced across groups, and aliquots should be assigned before the analyst sees group labels when feasible. For cultured cells, record passage, confluence, medium, time since perturbation and lysis conditions. For tissue, record dissection region, cold ischemia and storage duration; snap-frozen tissues often require mechanical homogenization. For plasma, record collection tube, anticoagulant, hemolysis, platelet contamination, time to centrifugation and freeze-thaw cycles. Fecal samples add matrix complexity and potential bacterial proteins; separation of microbial and host contributions may require a combined sequence database and explicit peptide taxonomic ambiguity handling.

Extract proteins with a buffer that solubilizes the desired compartment without undermining the planned enzyme digest or affinity capture. Strong detergent is helpful for membranes but can disrupt complexes in Co-IP experiments; these protocols cannot be substituted blindly. Determine protein concentration, reduce disulfides with a suitable reducing agent, alkylate free thiols, digest with trypsin or trypsin/Lys-C, and clean peptides to remove salts/detergents. In complex samples, bead-based S-Trap/SP3 or membrane-based FASP-style cleanup may be used according to compatibility, throughput and recovery requirements. An isotopic peptide or protein standard is useful when absolute recovery or targeted quantification is central to the study.

Before injection, use pooled reference digests and blanks appropriate to the instrument platform. A typical discovery workflow separates peptides by reversed-phase LC and records precursor MS1 scans and fragment MS2 spectra. MS1 estimates precursor m/z and intensity; MS2 contains sequence-informative fragment ions used for peptide identification. Trypsin tends to cleave after lysine and arginine except in sequence contexts that inhibit cleavage; missed cleavages and off-target cleavage should be included in the search policy, not treated as evidence of new proteins automatically.

2.3 Protein-centered interaction and combinatorial omics

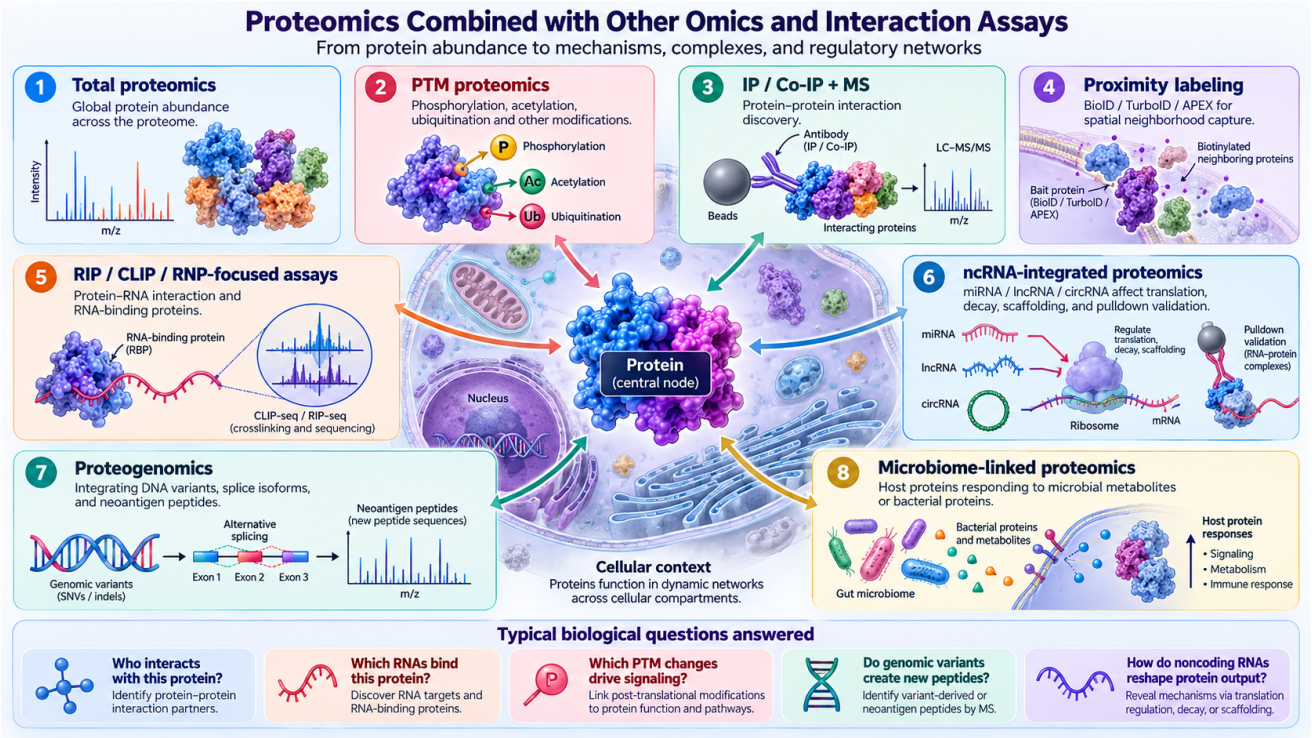


Figure 3. Proteomics does not only measure abundance. It can be integrated with interaction assays, PTM profiling, proteogenomics, RNA-binding assays, and microbiome-related functional studies to reveal mechanisms.

For undergraduates, it is useful to think of modern proteomics not as a single experiment but as a family of protein-centered strategies. Standard total proteomics asks how protein abundance changes between conditions. PTM proteomics asks how specific regulatory marks, especially phosphorylation, acetylation, ubiquitination, or glycosylation, reshape signaling and function. Interaction-focused proteomics asks which proteins are physically associated with a protein, RNA, membrane compartment, organelle, or biochemical niche. Together, these methods convert the proteome from a list of molecules into a mechanistic network.

2.3.1 Immunoprecipitation and Co-IP-MS: complex association

Scientific question: which proteins reside in the same complex as a bait under defined physiological conditions? Immunoprecipitation (IP) enriches a target antigen through an antibody. Co-IP seeks proteins that remain associated with the bait during lysis, capture and washing; mass spectrometry identifies candidate co-enriched preys. First validate bait expression, the antibody and the native localization of the complex. Select a lysis buffer that preserves interactions yet solubilizes the relevant compartment. Include a matched isotype or beads-only negative control and, when feasible, a genetic bait-null or epitope-tag-negative control. Collect input and unbound fractions to assess recovery. Elute, digest, identify and compare prey enrichment across bait and control conditions; use peptide evidence and abundance-aware statistical models rather than accepting every detected protein as a direct binder.

Experimental interpretation: enrichment in bait IP suggests biochemical co-association under the assay conditions; it does not prove direct physical contact, directionality or functional relevance. Nucleic acids may bridge proteins, so nuclease treatment can test RNA/DNA dependence, but nuclease treatment may also disrupt genuine assemblies. Reciprocal IP, endogenous tagging, recombinant binding or orthogonal imaging can strengthen mechanistic interpretation. Dynamic complexes may be lost during lysis; overly strong detergent or salt can generate false negatives, and promiscuous bead binders can generate false positives.

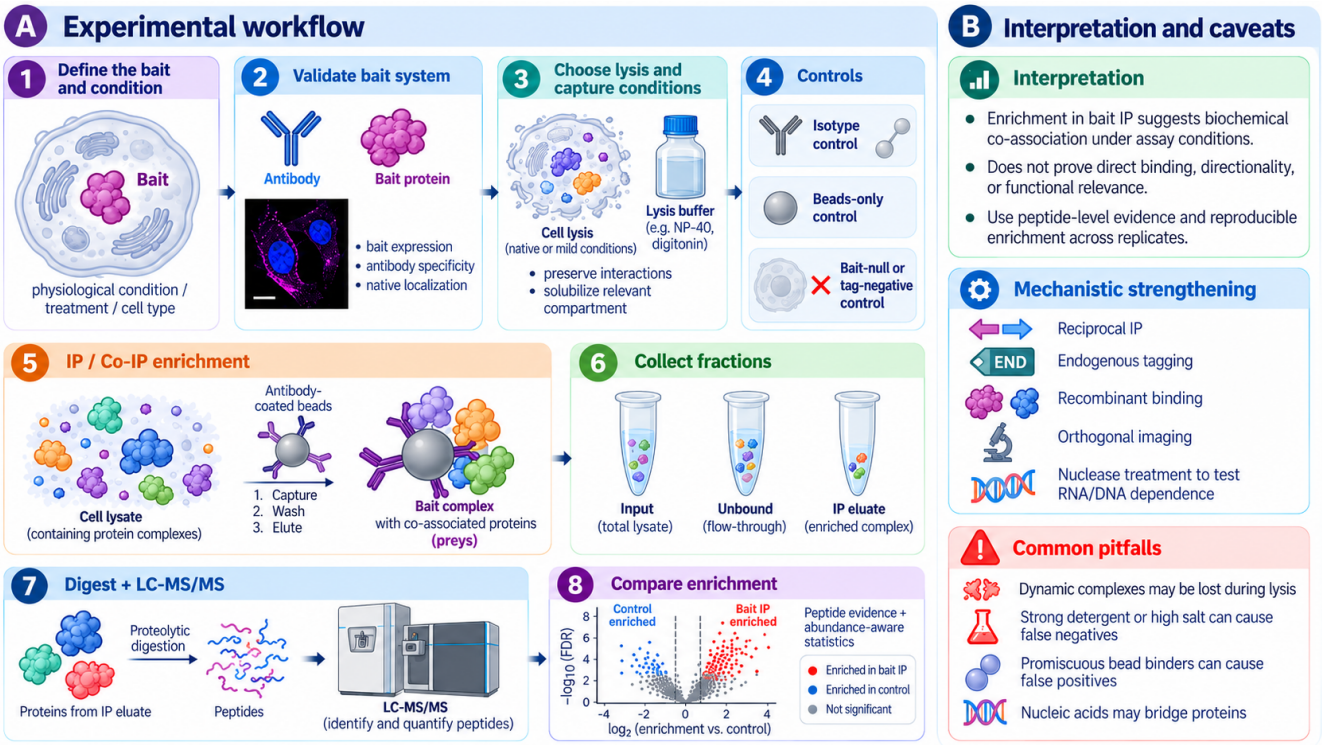


Figure 4. Immunoprecipitation (IP) enriches a bait antigen through an antibody, whereas Co-IP-MS identifies proteins that remain associated with the bait during lysis, capture, and washing.

2.3.2 Affinity purification-MS: tagged bait and interaction-network controls

Affinity purification-MS enriches a bait through an engineered epitope tag or other ligand-affinity system and is particularly useful when antibodies to the native target are weak or unavailable. Endogenous genome tagging avoids some overexpression artifacts, while tandem affinity approaches can improve specificity at the cost of potential loss of weak partners. The untagged cell line, empty-tag construct, localization check and protein-expression comparison are essential controls. An AP-MS interaction graph consists of candidate bait-prey associations and may include indirect members of large complexes; it should not be interpreted as a set of binary biophysical affinities. Differential AP-MS compares control and perturbed interactomes only after accounting for bait recovery and total prey-protein abundance.

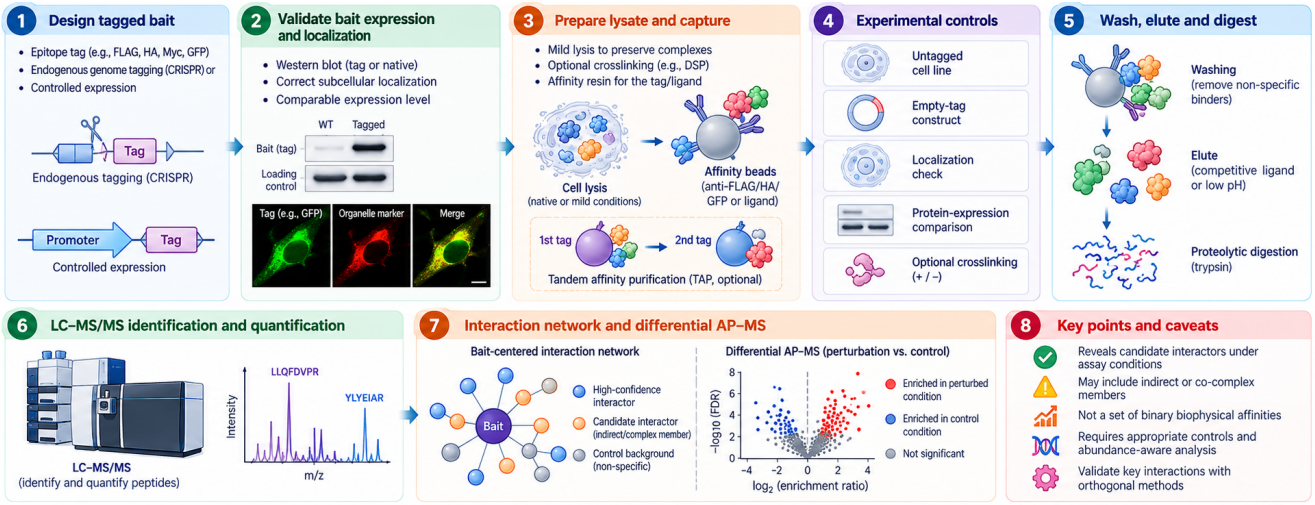


Figure 5. Affinity purification–mass spectrometry (AP-MS) for mapping bait-associated protein networks. (A) Experimental workflow. A genetically encoded epitope tag or ligand-affinity handle enables purification of the bait and co-enriched proteins. Endogenous tagging can reduce overexpression-related artifacts, whereas tandem affinity purification may improve specificity but lose weakly associated partners. Matched untagged-cell and empty-tag controls, together with bait-expression and localization checks, support the identification of candidate interactors by LC–MS/MS. (B) Data interpretation and validation. Candidate bait–prey associations are prioritized using reproducible enrichment relative to controls, with adjustments for bait recovery and total prey abundance when comparing conditions. AP-MS networks may contain indirect complex members; individual edges do not establish direct physical binding or interaction affinity. Reciprocal purification and orthogonal binding or functional assays can strengthen mechanistic conclusions.

2.3.3 BioID, TurboID, and APEX2: proximity rather than direct binding

Proximity-labeling enzymes are fused to a bait or directed to an organelle, where they covalently label nearby molecules. BioID-family enzymes use activated biotin intermediates, whereas APEX-family enzymes use a peroxidase reaction with a labeling substrate. TurboID is convenient for rapid biotin labeling; APEX2 can offer short labeling windows but requires careful control of the oxidant exposure and cellular stress. The labeled proteins are enriched with streptavidin and identified by LC-MS/MS. Optimize expression, subcellular targeting, labeling time, substrate and dose using pilot experiments; include enzyme-only or compartment-matched controls, no-substrate controls, biological replicates and microscopy validation of localization.

A proximity-labeling hit indicates occupancy of a local biochemical neighborhood during the labeling window, not necessarily stable complex membership or a direct interaction. Membrane topology, compartment boundaries, protein abundance and local diffusion affect labeling. Compare TurboID and APEX2 results only after acknowledging their distinct reaction chemistries and detection biases. A 2025 comparison demonstrated that the two enzymes can yield different compartment-associated profiles under the tested conditions [R4]. Strong mechanistic inference often uses orthogonal Co-IP, imaging, loss-of-function and a phenotype readout.

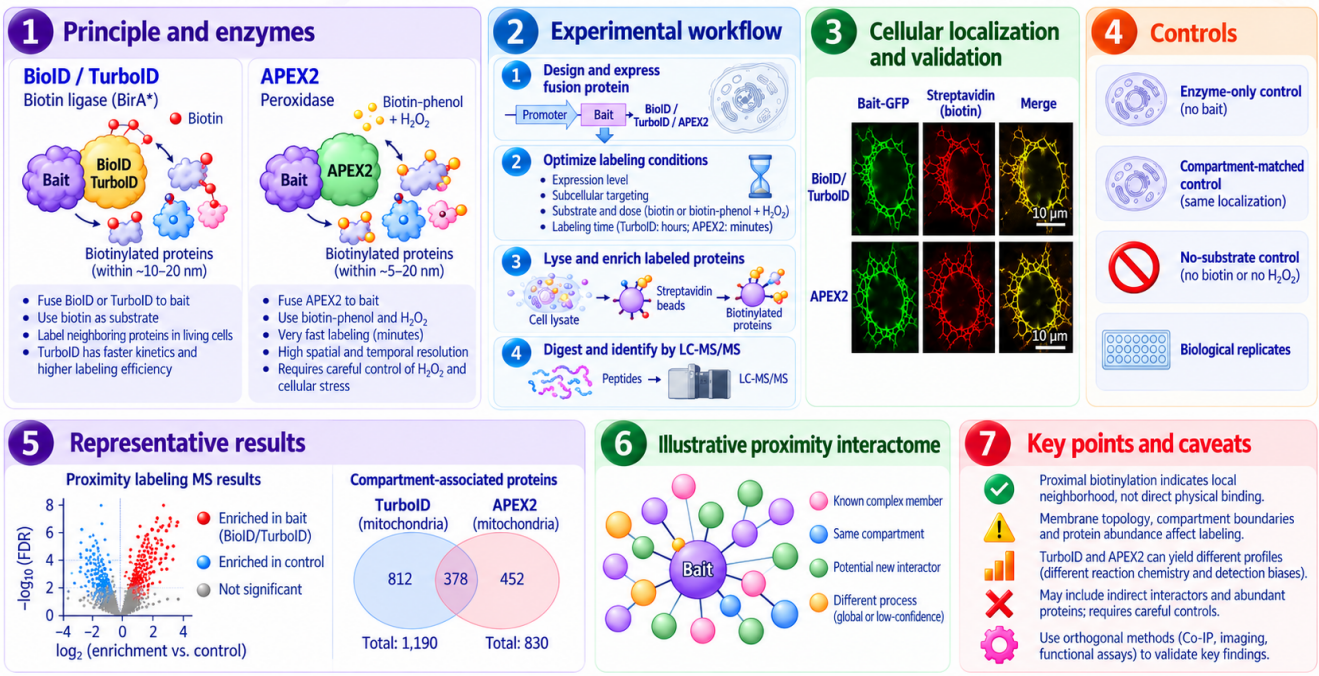


Figure 6. BioID, TurboID, and APEX2 proximity-labeling proteomics. (A) A proximity-labeling enzyme is fused to a bait protein or targeted to a defined subcellular compartment. BioID and TurboID use biotin-dependent labeling chemistry, whereas APEX2 catalyzes labeling through a peroxidase reaction. Nearby proteins are covalently labeled during a defined experimental window, enriched using streptavidin, digested, and identified by LC–MS/MS. (B) Experimental controls include compartment-matched enzyme controls, no-substrate controls, biological replicates, and verification of enzyme expression and localization. Labeling time, substrate concentration, and potential cellular stress require optimization. (C) Enriched proteins represent candidates within the bait's local biochemical neighborhood, not necessarily direct binding partners. Membrane topology, protein abundance, diffusion, and enzyme-specific labeling biases influence detection. Orthogonal Co-IP, imaging, loss-of-function experiments, and functional assays can strengthen mechanistic interpretation.

2.3.4 PTM proteomics: enrichment, localization, and protein abundance

Phosphoproteomics often uses metal oxide affinity chromatography (for example TiO2) or immobilized metal affinity chromatography after proteolysis; enrichment of phosphopeptides is necessary because most modified species are low abundance relative to their unmodified counterparts. For ubiquitination, antibodies recognizing lysine di-glycine remnants after appropriate digestion can enrich ubiquitin-associated sites, but the biological origin of every di-glycine tag should not be oversimplified. Acetyl-lysine and glycopeptide enrichments require their own chemistry and quality controls. Record peptide localization confidence, phosphosite multiplicity, neutral-loss patterns where relevant, parent-protein abundance, false localization and batch-specific enrichment efficiency.

Design an activation experiment with a time course rather than a single late endpoint if phosphorylation is expected to change rapidly. Integrate phosphosite dynamics with total proteomics, kinase substrate databases and a directly measured functional assay. A signaling model becomes more credible when total receptor abundance, site-specific phosphorylation, downstream target activation and a perturbation-response phenotype agree without contradicting basic sample QC.

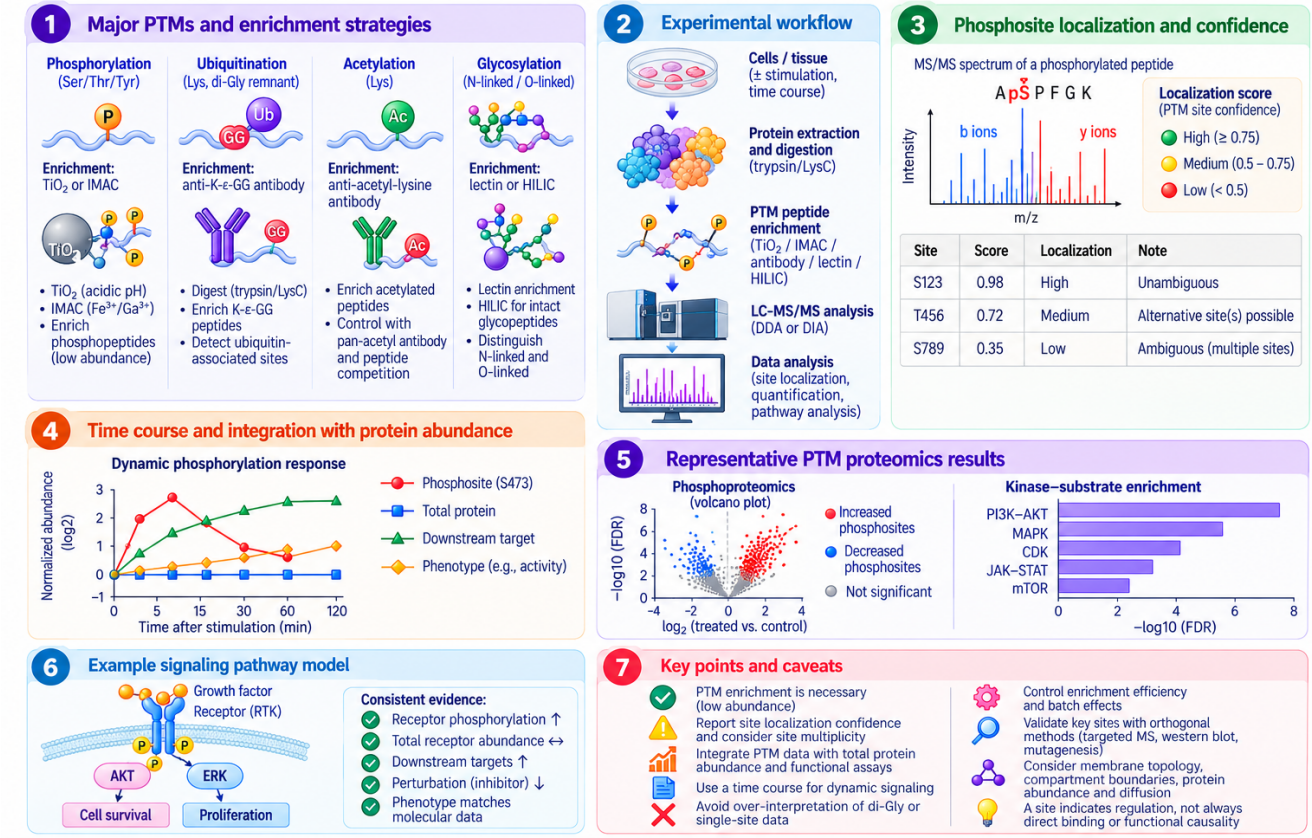


Figure 7. PTM proteomics: enrichment, site localization, and protein-abundance-aware interpretation. (A) Proteins collected across a defined stimulation time course are extracted, digested, and divided into total-proteome and PTM-enrichment workflows. Phosphopeptides are enriched using TiO_2 or IMAC; ubiquitin-associated di-glycine remnants, acetylated peptides, and glycopeptides require modification-specific enrichment strategies. (B) LC–MS/MS identifies and quantifies modified peptides. Site-localization confidence, peptide identification FDR, enrichment efficiency, batch effects, and parent-protein abundance are assessed before comparing modification levels. A change in phosphopeptide abundance may reflect altered protein abundance rather than a change in phosphorylation stoichiometry. (C) Time-resolved phosphosite profiles are integrated with total proteomics, kinase–substrate annotations, downstream signaling measurements, and perturbation-response assays to develop and test mechanistic hypotheses.

2.3.5 RNA-protein interactomics: RIP, CLIP, and RNA pull-down-MS

RIP/RIP-seq pulls down an RNA-binding protein and detects associated RNAs by qPCR or sequencing; it is appropriate for asking which RNA species are enriched with a selected RBP under the tested lysis conditions. UV-CLIP and variants such as eCLIP add crosslinking and sequencing-based analysis of RNA-binding regions and can more directly constrain RNA-RBP contact locations; read mapping, crosslink biases, RNase digestion and background control must be interpreted carefully. These methods do not themselves produce a global protein-abundance table. Conversely, an RNA pull-down with labeled bait RNA followed by MS asks which proteins bind or associate with that RNA; length-matched and sequence-scrambled controls, antisense probes, subcellular localization and RNA structure matter.

For a lncRNA-mediated mechanism, pair RNA pull-down-MS with candidate-protein RIP/eCLIP, characterize the protein complex by Co-IP-MS and measure downstream transcript, protein and metabolite changes after RNA loss and rescue. For miRNA mechanisms, seed-site reporter control, AGO-related RIP/CLIP context and proteomic confirmation of target repression answer complementary questions. A 2025 Nature study combined an irCLIP strategy with MS to identify RNA-associated protein assemblies and a sequential Re-CLIP workflow to interrogate co-occupied RNAs [R5]; this is a concrete example of crossing the boundary between RNA interactome mapping and proteomics rather than labeling every RIP experiment “proteomics.” ncRNA-integrated proteomics: miRNAs can regulate translation or degradation of specific mRNAs; lncRNAs can scaffold proteins, recruit chromatin modifiers, or modulate signaling; circRNAs can sponge miRNAs or bind proteins. Integrating ncRNA data with proteomics helps students understand why RNA abundance alone may fail to predict protein output.

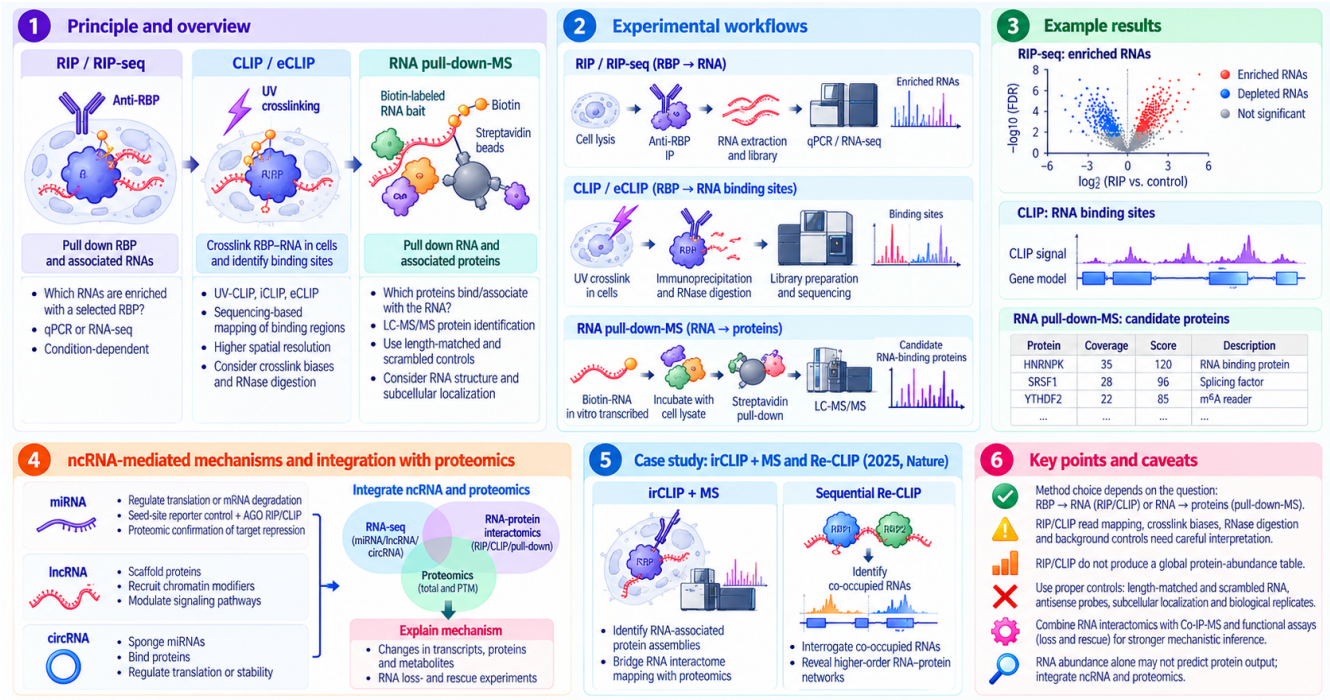


Figure 8. RNA–protein interactomics: RIP, CLIP, and RNA pull-down–MS. (1–3) RIP/RIP-seq identifies RNAs associated with a selected RNA-binding protein (RBP), CLIP/eCLIP maps RNA-binding regions, and RNA pull-down–MS identifies proteins associated with a selected RNA. (4) Integration of these assays with transcriptomics, proteomics, and functional perturbations enables investigation of miRNA-, lncRNA-, and circRNA-mediated regulation. (5) Advanced irCLIP–MS and sequential Re-CLIP approaches reveal RNA-associated protein assemblies and co-occupied RNAs. (6) Appropriate controls, biological replicates, and orthogonal validation are essential, as RNA–protein association alone does not establish direct binding or functional causality.

2.3.6 Proteogenomics and immunopeptidomics

Integrate sample-matched WGS/WES, RNA-seq splice information and mass spectra to build a carefully constrained personalized peptide search space. A larger database permits more hypotheses but increases the multiple-testing burden and the risk of incorrect peptide assignments; use suitable decoy/entrapment strategies, peptide uniqueness checks, manual spectrum review for critical claims and orthogonal verification of unusual peptides. Immunopeptidomics enriches HLA-presented peptides, which are generally not generated by ordinary trypsin digestion; presentation is not equivalent to immunogenicity. This approach connects genomic variants and noncanonical translated RNA products to actual peptide presentation. The NeoDisc study integrates immunopeptidomics, genomics and transcriptomics for cancer antigen discovery [R6]. Its translational lesson is the use of orthogonal data to reduce the gap between predicted candidate antigens and observed presented peptides. Undergraduate readers should distinguish a genome-predicted mutation, an RNA-expressed sequence, an MS-supported peptide and a T-cell-functional antigen as four different levels of evidence. A useful teaching rule is this: when the biological question starts with “who binds,” “what complex forms,” “which PTM drives signaling,” or “how does a noncoding RNA reshape protein output,” students are no longer in the territory of simple bulk abundance measurement. They are designing a multi-layer mechanism experiment.

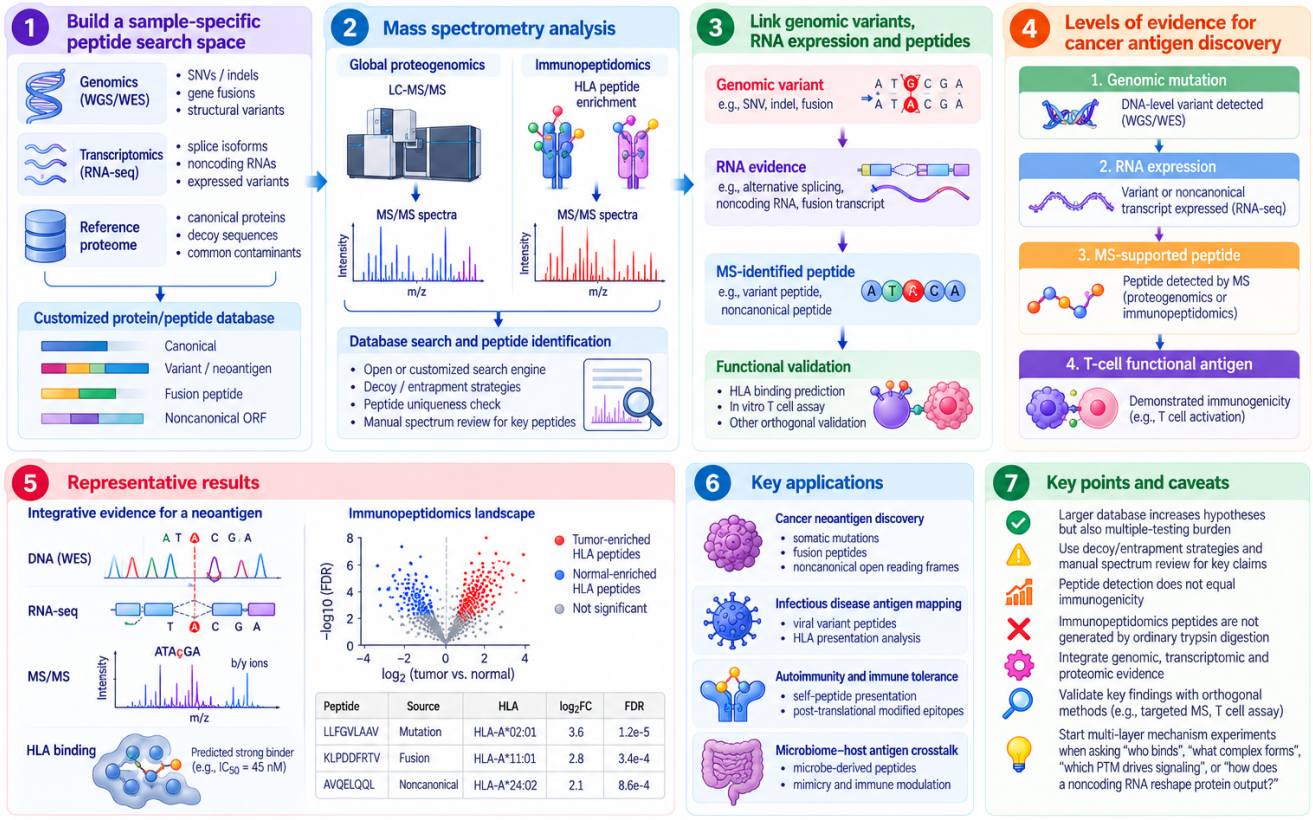


Figure 9. Proteogenomics and immunopeptidomics: from genomic variation to functional antigen validation. (A) Matched WGS/WES and RNA-seq data are used to identify sequence variants, expressed splice junctions, and candidate noncanonical translation products. These findings inform a constrained, sample-specific peptide search database for LC–MS/MS analysis. (B) Immunopeptidomics uses HLA immunoaffinity purification to enrich naturally presented peptides, which are analyzed without routine trypsin digestion. Candidate peptide assignments require appropriate false-discovery control, sequence-uniqueness assessment, and additional verification for unusual or clinically important findings. (C) Four evidence levels must be distinguished: a genome-predicted variant, an RNA-expressed sequence, an MS-supported HLA-presented peptide, and an antigen demonstrated to elicit a functional T-cell response. HLA presentation alone does not establish immunogenicity.

2.4 Frontier: single-cell, spatial, and structure-guided proteomics

Single-cell protein molecules cannot be amplified like DNA, so the central challenge is sensitivity and sample loss. Miniaturized preparation, high-sensitivity chromatography, newer mass spectrometers, intelligent DIA windows and careful carrier-channel design have expanded sensitivity. In a 2025 Nature Methods Chip-Tip study, the authors reported identification of more than 5,000 proteins from individual HeLa cells under their optimized experimental conditions [R2]; this is an example of method performance in a defined system, not a routine expectation for all cell types or laboratories. Single-cell studies need plate position/batch monitoring, empty wells, doublet control, cell size and cell-cycle covariates, and validation of rare-cell phenotypes.

Spatial proteomics can mean antibody-based imaging (CODEX/PhenoCycler-type cycles, imaging mass cytometry, MIBI) or mass-spectrometry measurement of spatially selected cells or microregions (deep visual proteomics). Antibody panels provide location and cellular neighborhoods but target a restricted set of antigens; LC-MS-based microdissection can extend molecular breadth while requiring tissue morphology registration and careful sampling. In 2024 Nature Methods selected spatial proteomics as its Method of the Year [R3], underscoring the demand for spatial context in tissue biology. These technologies become especially informative when cross-registered to spatial transcriptomics and metabolite imaging, but matched coordinates and compatible resolution are prerequisites.

AI-based protein structure models add a different dimension: AlphaFold-type predictions can generate candidate structures and complexes for further hypothesis generation, not measured protein abundance or verified binding. Interpret confidence metrics, construct design, flexible regions and relevant cofactors; predicted ligand or partner binding must be checked against biochemical or cellular evidence. For protein engineering, structure prediction can help prioritize mutations before a Design-Build-Test-Learn cycle, while proteomics quantifies expression and stability and metabolomics measures actual product formation.

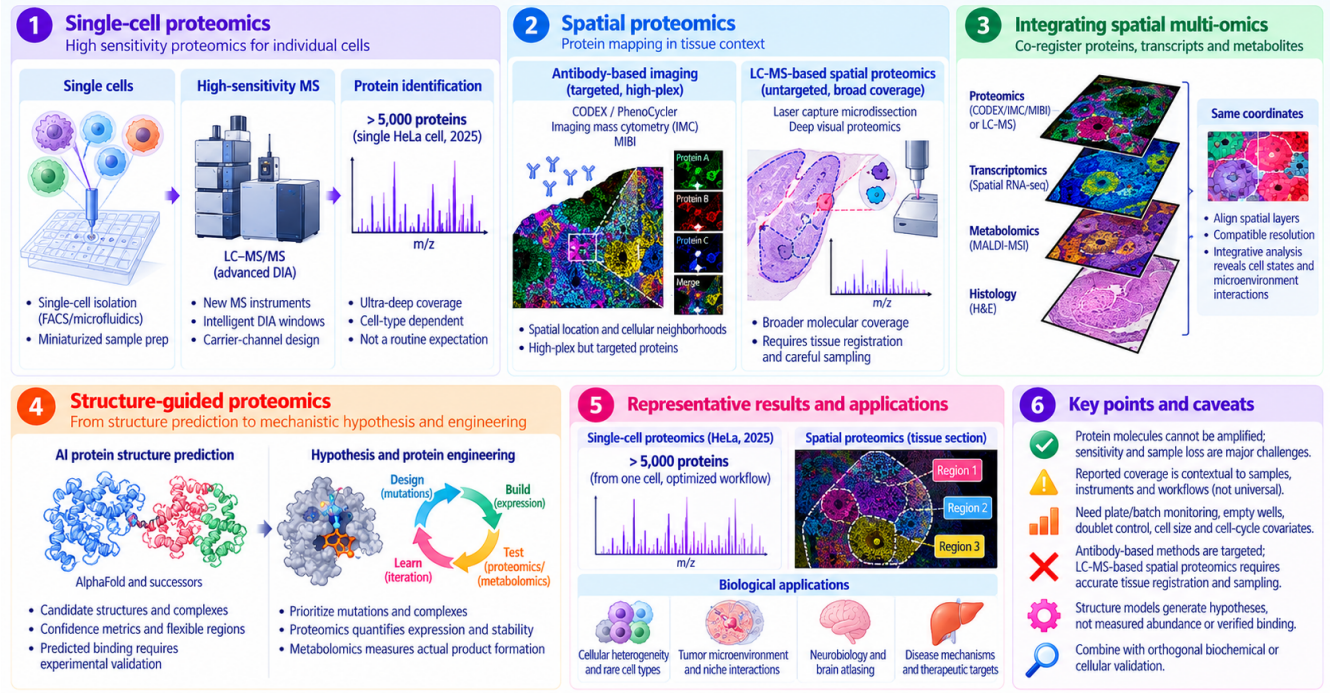


Figure 10. Frontier proteomics: single-cell, spatial, and structure-guided approaches. Single-cell proteomics combines miniaturized preparation and high-sensitivity MS to quantify proteins from individual cells. Spatial proteomics maps protein distributions in tissue using multiplexed imaging or LC–MS-based microdissection and can be integrated with spatial transcriptomics and metabolite imaging. Structure-guided proteomics uses AI-predicted protein structures to prioritize mechanistic hypotheses and engineering designs, which must be experimentally validated. Reported coverage and performance remain dependent on specimen type, instrumentation, workflow, and appropriate quality control.

2.5 Recent advances: human tissue proteome atlas and bacterial single-cell proteomics (2026)

Yue and colleagues reported a spatially resolved quantitative human proteome resource obtained by DIA across 2,856 samples, including 58 major tissue types, 251 tissue subtypes and 25 carcinomas, with more than 13,000 proteins profiled [R14]. This is a reference atlas for formulating cell-, tissue- and disease-context hypotheses, not a claim that the same 13,000 proteins are reliably measurable in every single sample. In synthetic biology and microbiome research, a September 2026 Nature Communications report introduces bacSCP to investigate bacterial heterogeneity through single-cell proteomics [R15]. Students should ask what cell-isolation, low-biomass controls, peptide identifiability and microbial species assignment are needed before extending this system to a mixed consortium.

3 | What instrument output must be checked before you trust a result

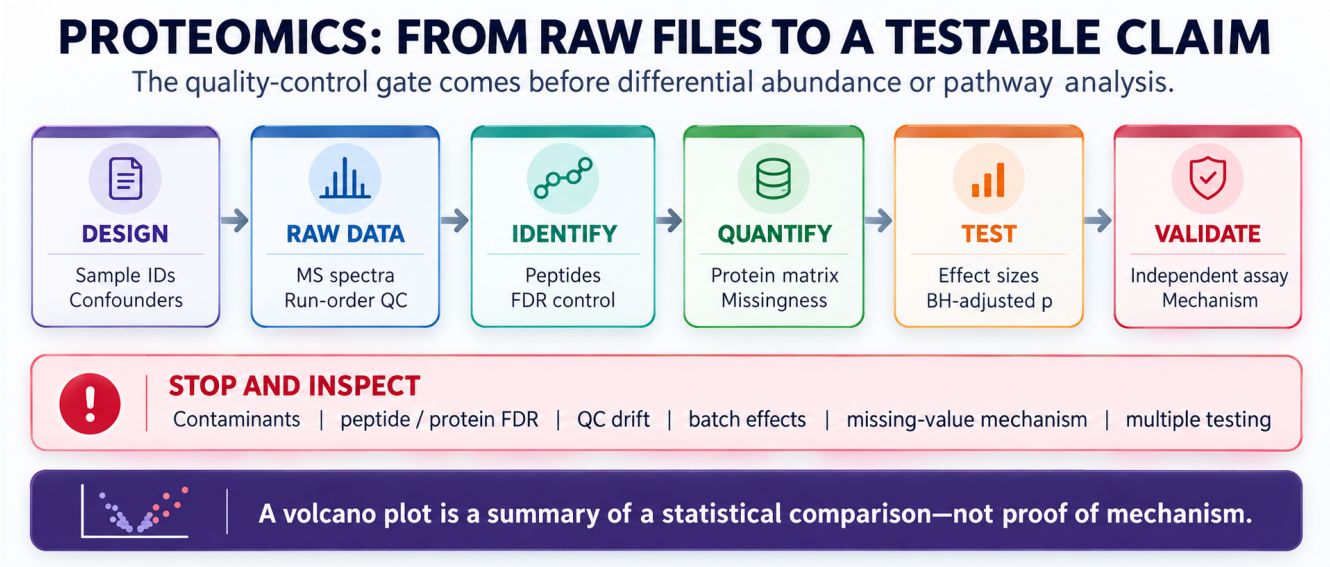


Figure 11. Proteomics QC logic. Identification and inference are meaningful only after sample design, raw-data inspection, missingness assessment, and contamination control.

Students often jump directly from a software output file to a volcano plot. This is scientifically unsafe. In proteomics, the raw output after instrument acquisition contains spectra and chromatographic information that must pass several quality-control gates before any biological comparison is credible. In metabolomics, the same is true for feature tables, retention-time drift, blanks, pooled QCs, and annotation evidence.

3.1 Proteomics QC checklist

- Design-level QC: confirm sample identifiers, group labels, randomization, batch structure, sex/age/covariate balance, and whether any sample swaps are plausible.
- Raw-signal QC: inspect total ion current, base-peak chromatograms, injection stability, retention-time alignment, precursor-intensity distribution, and instrument drift across run order.
- Identification QC: review PSM, peptide, and protein counts; verify target-decoy false discovery control; inspect contaminant and reverse-hit flags.
- Quantification QC: quantify missingness per protein and per sample, check intensity distributions, pairwise sample correlations, coefficient of variation across technical or pooled controls, and obvious outliers.
- Biology-aware QC: evaluate whether observed variation is dominated by batch, tissue quality, storage time, or preparation artifact rather than by the intended biological contrast.

Only after these checks should students perform differential abundance testing using an appropriate design matrix. A statistically significant protein is still not a mechanistic conclusion: it is one piece of

evidence to be interpreted alongside known function, pathway context, orthogonal assays, and experimental perturbation.

3.1.1 Instrument-level QC, protein inference, and differential testing

Raw-data QC starts with LC gradient stability, total ion current, retention-time drift, injection carryover, mass accuracy, precursor charge-state distribution and the abundance of standard peptides. At the identification stage, distinguish peptide-spectrum match (PSM), peptide and protein-group FDR. A 1% peptide identification FDR does not mean that a between-group protein-abundance result has $p = 0.01$. Target-decoy competition estimates identification error; differential-testing p values evaluate a completely different null hypothesis. Protein inference must account for peptides shared across proteins. In a metaproteomics experiment, many peptides may be shared among related bacterial species, limiting taxonomic specificity.

Before statistical testing, remove confirmed contaminants and explicit decoy or site-only identifications as appropriate; visualize number of identified features per run, log-intensity distributions, missing-value patterns, replicate correlations and PCA. A missing intensity can reflect abundance below detection, stochastic sampling or data-processing effects; a zero in a vendor LFQ table is not an experimentally measured zero concentration. Log-transform positive intensities, justify normalization against expected global shifts and decide whether missing values can be modeled without imputation. Use limma, MSstats, proDA or other method compatible with the actual matrix and design. Record fold-change thresholds separately from multiplicity-adjusted significance and inspect sensitivity to filtering and imputation choices.

3.1.2 Interpreting volcano plots, enrichment, and phosphosite results

A volcano plot contains effect size (typically log2 fold change) and statistical evidence (often -log10 adjusted p); the horizontal axis is not protein activity, and the vertical axis is not mechanistic importance. An intensity heatmap may be row-standardized, so its color scale represents relative within-protein patterns rather than absolute protein abundance. A GO/KEGG over-representation test requires a background consisting of proteins that were measurable and testable in the study; using the entire genome as the default background can distort enrichment. A pathway dot plot may encode overlap ratio, count or adjusted p in its axes/color/size; students must read the figure legend before drawing conclusions.

For phosphoproteomics, compare a changed phosphosite with the abundance of its parent protein wherever possible. An apparent twofold increase in a phosphopeptide may reflect more protein rather than higher site occupancy. Site-localization probability and interference must be reported; enrichment of a kinase motif is not direct proof that the named kinase phosphorylated the site. A defensible sequence is pathway-consistent phosphosite changes, candidate kinase analysis, targeted phosphosite verification, kinase inhibition or genetic perturbation, and a functional endpoint.

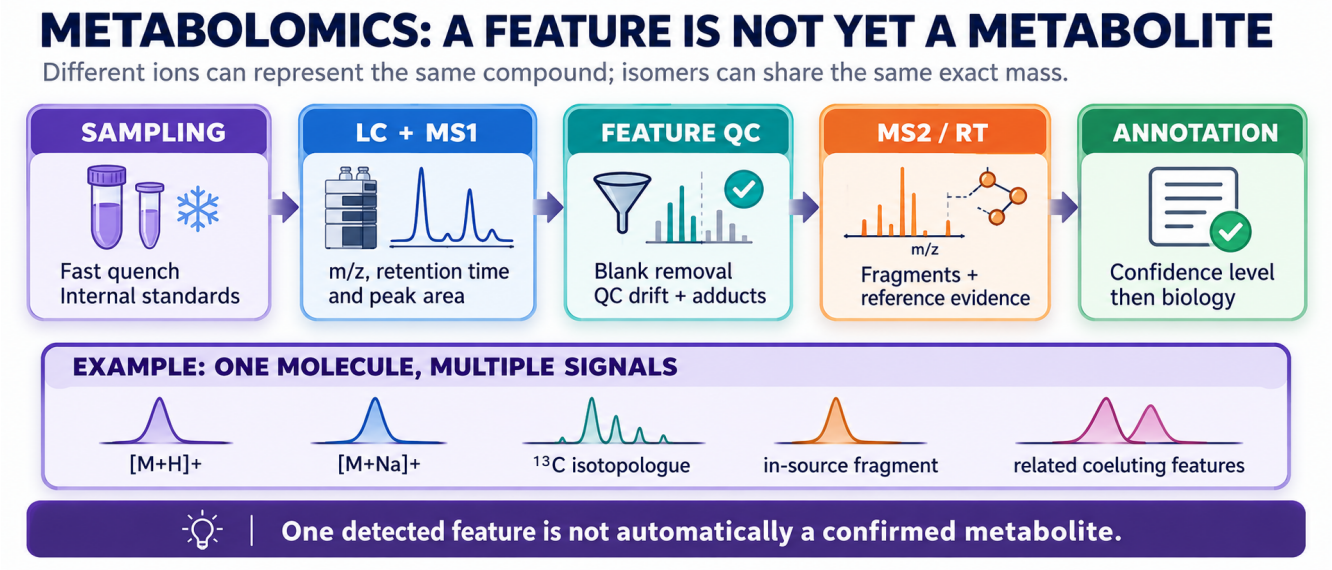


Figure 12. Metabolomics QC and annotation logic. A detected feature is not automatically a confirmed metabolite; adducts, isotopes, fragments, and coeluting signals must be filtered before biological naming.

3.2 Metabolomics QC checklist

- Sampling QC: confirm rapid quenching, consistent storage, extraction timing, and addition of internal standards at the earliest possible step.
- Blank and carryover QC: remove features enriched in blanks or dominated by solvent/background contamination.
- Pooled-QC monitoring: inspect pooled QC clustering, retention-time drift, peak-area stability, and relative standard deviation across QCs.
- Feature-level QC: remove features with excessive missingness, poor reproducibility, or strong run-order drift; group adducts, isotopes, and in-source fragments where appropriate.
- Annotation QC: explicitly distinguish confidence levels. Mass-only matching is weaker than RT + MS2 + authentic-standard confirmation. Interpret chemistry only after reporting the evidence level.

This is a crucial teaching point for microbiome-related metabolomics. Because many host-microbe mediators—such as bile acids, indoles, and short-chain fatty acids—exist in chemically related families, misannotation can produce biologically attractive but false stories. Students should therefore learn to report not just the metabolite name, but also how confident the identification is.

4 | Metabolomics: from feature tables to functional chemistry

The metabolome is the small-molecule complement of a biological system. Compared with proteomics, metabolomics sits closer to phenotype because metabolites directly reflect pathway output, nutrient input, host-microbe exchange, drug exposure, and redox or energetic state. However, this proximity to phenotype comes with a challenge: many measured features are difficult to annotate, different ion species

can derive from the same compound, and a metabolite pool size does not automatically reveal pathway direction or flux.

Students should separate three major levels of metabolomics. First, untargeted metabolomics provides global feature discovery and is ideal for finding candidate biomarkers or pathway shifts. Second, targeted metabolomics quantifies a defined panel using authentic standards and, ideally, stable-isotope internal standards. Third, isotope tracing and flux-oriented studies ask not simply how much of a metabolite is present, but where it came from and how quickly it is turning over. This distinction becomes especially important when students transition from descriptive biology to mechanistic reasoning.

Week 7 should also reinforce the host-microbe context introduced in Week 6. Many of the most important microbiome-linked molecules are best handled through targeted metabolomics after an untargeted discovery phase. Examples include bile acids, tryptophan derivatives, SCFAs, branched-chain amino acids, trimethylamine-N-oxide-related metabolites, and lipid mediators. A typical teaching workflow is therefore: first detect a chemical difference globally, then confirm identity and concentration in a targeted assay, then integrate that result with microbial genes, host proteins, and phenotype.

4.1 Chemical diversity, metabolite pools, and analytical scope

An untargeted experiment detects ion features in a defined analytical window; it does not measure every metabolite in the sample. Coverage depends on sample matrix, chromatography, ionization, polarity, instrument range and extraction chemistry. Targeted metabolomics quantifies defined compounds with appropriate standards and calibration. Both can use LC-MS; GC-MS is advantageous for many volatile or derivatizable metabolites; NMR offers strong structural information and quantitative robustness for sufficiently abundant compounds; imaging MS adds location at the expense of a different set of constraints. Lipidomics often requires class-specific extraction and reporting of isomer ambiguity. A single retention-time / m/z feature can be an adduct, isotopic feature or fragment, and a single compound can generate many features.

The metabolite pool reflects net production, consumption, transport and dilution. Increased citrate concentration does not prove increased TCA-cycle flux; a metabolite may accumulate because its downstream consumption is blocked. Stable-isotope tracing estimates precursor contributions and labeling dynamics, while formal flux inference usually needs kinetic modeling, measured uptake/secretion and appropriate isotopologue correction. This distinction between concentration and flux is particularly important in cancer metabolism, aging, microbial fermentation and synthetic pathway engineering.

4.2 Matrix-specific collection and sample preparation

Matrix / analytical target	Preparation principle	Important control / caveat
Plasma/serum polar metabolites	Cold protein precipitation with validated solvent composition	Fasting, hemolysis, medications, internal standards, drying losses

Cellular central carbon	Rapid quench and cold solvent extraction; normalize to cell biomass	Time-to-quench and medium carryover can dominate true biology
Tissue metabolome	Rapid freeze/clamp; homogenize cold; consider tissue compartment	Post-dissection ischemia and spatial heterogeneity
Fecal SCFAs	Validated extraction plus derivatization where assay requires	Wet/dry mass, pH, storage and non-microbial host sources
Lipidomics	Validated monophasic/biphasic extraction such as MTBE-based workflow	Lipid-class recovery, oxidized lipids, matrix-specific standards
Fermentation supernatant	Clarify cells rapidly and sample time course with volume correction	Evaporation, medium background, growth phase, extraction recovery

Internal standards should usually be introduced early enough to report sample-processing variation; one isotope-labeled standard does not correct the behavior of every chemical class. Extraction blanks reveal environmental background, pooled QC samples monitor analytical repeatability, dilution-series QC can test feature response behavior and repeated reference standards support long-run instrument checks. Freeze-thaw, oxygen exposure, solvent evaporation and microbial metabolism after collection can change apparent concentrations. The analytical protocol should report the matrix, extraction solvent, sample-to-solvent ratio, quench interval, centrifugation and storage conditions, but no single recipe is optimal for all metabolite classes.

4.3 LC-MS, GC-MS, NMR, ion mobility, and imaging platforms

Reversed-phase LC tends to retain hydrophobic molecules such as many lipids and bile acids, whereas HILIC can support the retention of highly polar metabolites such as nucleotides and amino acids. Electrospray positive and negative ionization expand chemical coverage but introduce different adduct and ion-suppression patterns. High-resolution mass spectrometers can constrain elemental formulas when isotope patterns and precursor mass are reliable; collision-induced fragmentation provides further structural evidence. Ion mobility supplies an additional separation axis and collision-cross-section information under validated conditions. MZmine 3 supports LC-MS, GC-MS, MS imaging and ion-mobility-associated feature processing in its documented workflows [R7].

In GC-MS, derivatization can improve volatility of organic acids or sugars but can also generate additional analyte derivatives and compound-specific response changes. NMR offers reproducible spectra and often stronger absolute or semi-absolute quantitation for major metabolites, but limited sensitivity for low-abundance analytes. MALDI-based metabolite imaging retains tissue context, while matrix effects, spatial resolution and isomeric ambiguity demand local validation and careful interpretation of displayed images.

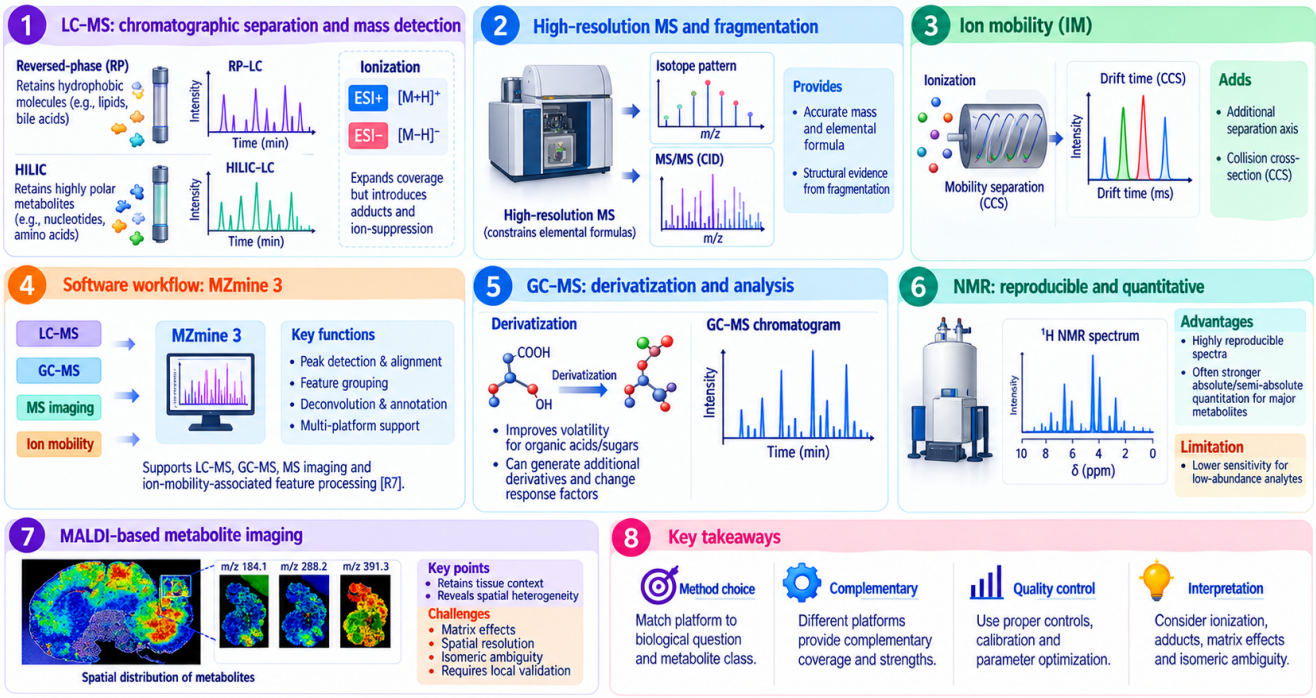


Figure 13. Major analytical platforms for metabolomics. LC–MS, GC–MS, NMR, ion mobility, and MALDI imaging provide complementary chemical coverage, structural information, quantification, and spatial context. Platform selection should match metabolite chemistry and the biological question while accounting for ionization and matrix effects, derivatization artifacts, sensitivity, spatial resolution, isomeric ambiguity, and annotation confidence.

4.4 Feature detection, alignment, and evidence-graded identification

A reproducible LC-MS analysis converts raw vendor files to a suitable open format (often mzML) while retaining polarity, centroid/profile status and required scan metadata. A peak picker detects chromatographic features, retention-time alignment maps the same analytes across runs, grouping establishes a sample-by-feature table, and targeted gap filling seeks expected signals where the initial picker failed. Next, remove features dominated by blanks or unstable in pooled QCs, inspect run-order drift, group plausible adducts and isotopes, and export quantitative data together with a separate annotation table. XCMS and MZmine are established alternatives; settings such as ppm, expected peak width and retention-time correction must follow chromatographic behavior rather than a copied universal parameter list. Annotation confidence matters. An exact mass alone is compatible with multiple formulas, adducts or isomers; an accurate formula does not provide unique molecular identity. An authentic standard under matched chromatography with consistent precursor, retention time and MS2 fragmentation supports a substantially stronger claim. Library-matched MS2 and in-silico ranked candidates should remain appropriately qualified, while molecular-class assignments (for example through CANOPUS) are weaker than a uniquely confirmed structure. SIRIUS/CSI:FingerID, GNPS-style molecular networking and related ML models can prioritize candidates but cannot transform mass-only evidence into confirmed identity. Compound structures produced by AI must be checked against chemistry and reference spectra before a manuscript or clinical claim.

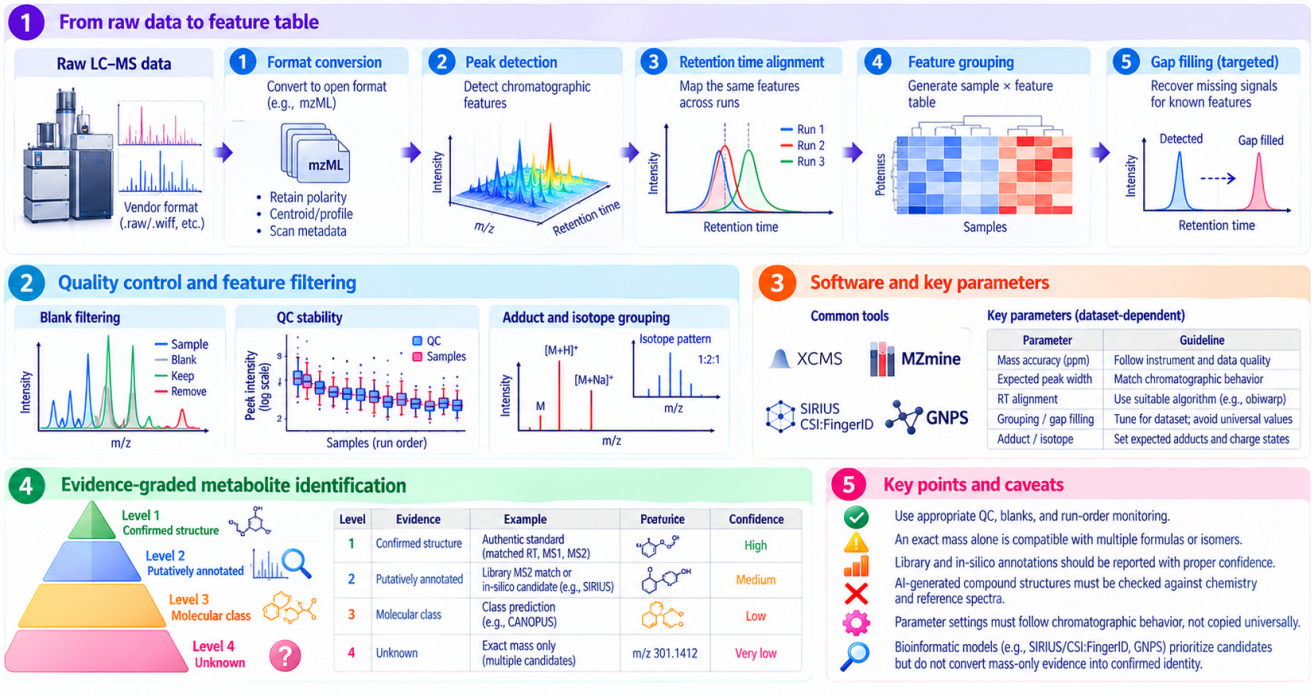


Figure 14. Feature detection, alignment, and evidence-graded metabolite identification. Raw LC–MS data are converted, peak-picked, aligned, grouped, gap-filled, and filtered using blanks, pooled QCs, run-order behavior, adducts, and isotopes to generate a quantitative feature table. Metabolite identities should be reported according to evidence strength, progressing from mass- or formula-based candidates to MS2-supported annotations and, at the highest confidence, authentic-standard confirmation under matched chromatographic conditions; computational tools can prioritize candidates but do not substitute for experimental evidence.

4.5 Targeted metabolomics: calibration and analytical validity

Design a targeted panel around a specific physiological pathway and expected concentration range. Select standards for each analyte when available, ideally alongside stable-isotope internal standards with similar retention and ionization behavior. Establish calibration curves across the physiological and pathological range; report matrix-matched calibration, blank response, analyte carryover, extraction recovery, matrix effects, dilution integrity, limit of detection, lower limit of quantification and within-/between-run precision. Check chromatographic separation for critical isomers such as bile acids or amino-acid enantiomers if the biological claim depends on them. A targeted concentration can only be called “absolute” when calibrated to a known standard with an explicit unit and validated range. For SCFAs, GC-MS or validated LC-based derivatization methods may be appropriate depending on matrix and throughput. For bile acids, LC-MS/MS panels should differentiate primary/secondary and conjugated species where relevant. For tryptophan metabolites, chemical specificity matters because biologically distinct indoles can have related masses and overlapping signals. In clinical plasma studies, targeted validation should use independent samples and calibrators/QCs distributed across all batches.

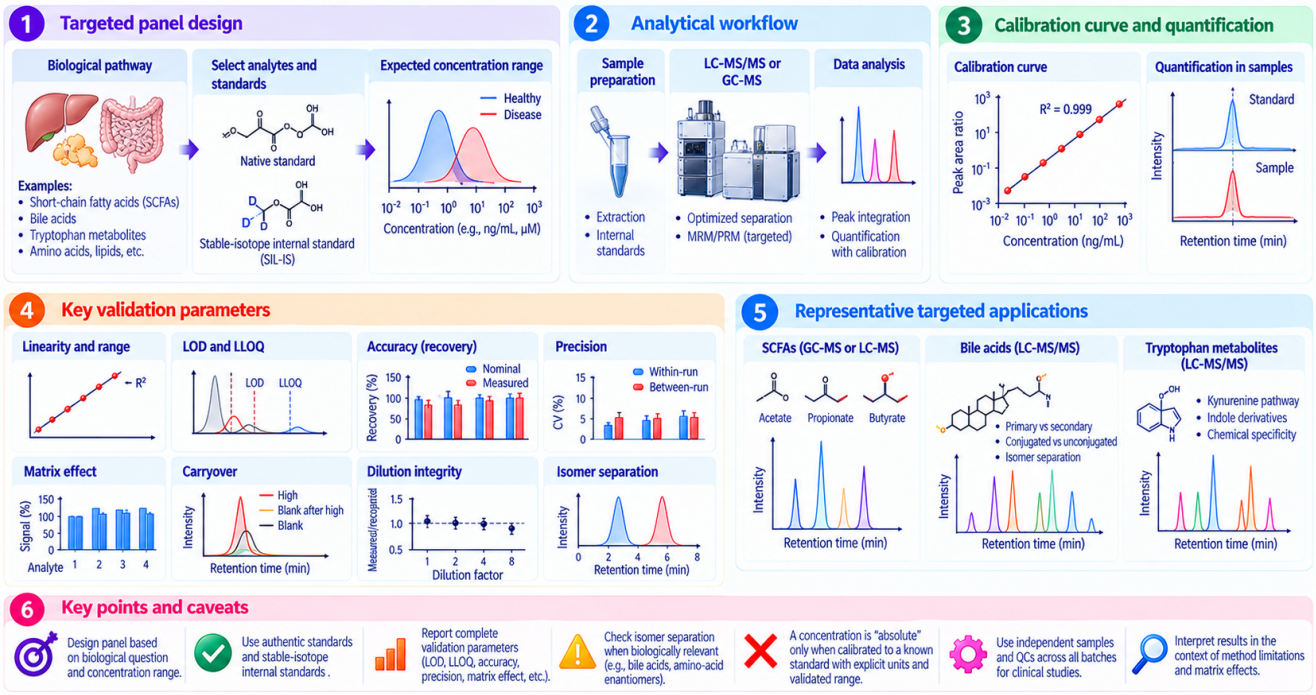


Figure 15. Targeted metabolomics: calibration and analytical validity. Targeted assays quantify predefined metabolites using authentic standards and, ideally, stable-isotope internal standards, with validated calibration, sensitivity, recovery, precision, matrix effects, carryover, dilution integrity, and isomer separation. Reliable absolute quantification requires explicit units, a validated analytical range, and batch-wide QC with independent validation samples.

4.6 Stable-isotope tracing, metabolic flux, and spatial metabolomics

A labeling experiment begins with an explicit atom-level question. For example, labeled glucose can test glucose-derived carbon incorporation into lactate or fatty acids; labeled acetate may interrogate acetyl-CoA contributions, and a labeled microbial precursor may help distinguish microbial conversion from host-only synthesis. Draw the carbon map, choose a tracer and sampling schedule, measure pool sizes as well as isotopologue fractions, subtract natural isotope abundance and consider precursor enrichment. A labeling fraction is not equivalent to flux without further modeling. Choose time points spanning the relevant kinetic regime and report the mathematical assumptions behind any flux estimate.

Spatial metabolomics associates ions with pixels or selected cell populations in tissue. Its interpretation requires tissue integrity, co-registration to histology, careful correction of intensity gradients and chemical identification evidence. A 2024 Nature Methods study combined MALDI imaging metabolomics with imaging mass cytometry on the same tissue section and analyzed metabolic heterogeneity among cell populations [R8]. A 2024 Nature Metabolism study used spatial single-cell isotope tracing to study glucose-derived fatty-acid synthesis heterogeneity in cancer systems [R9]. The important conceptual advance is the ability to ask "which cell type, where, and using which precursor?" rather than merely "is this metabolite higher in the whole tissue?"

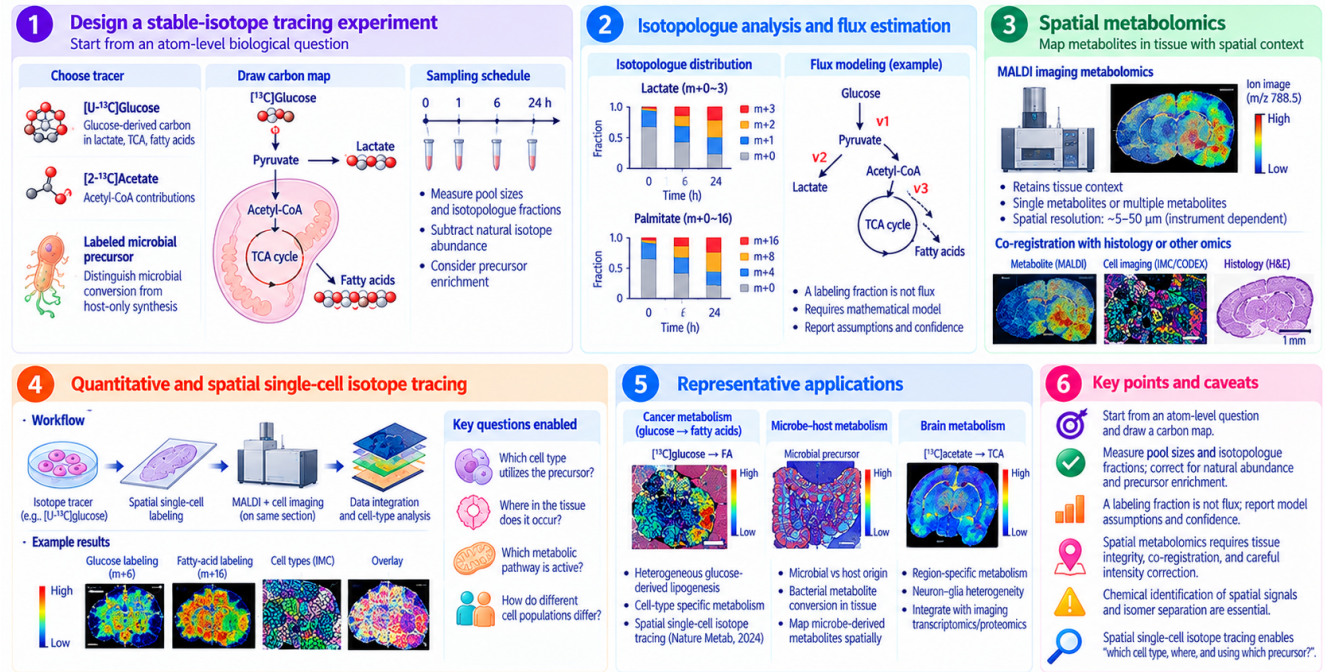


Figure 16. Stable-isotope tracing, metabolic flux, and spatial metabolomics. Stable-isotope tracing follows labeled atoms through metabolic pathways using defined tracers, kinetic sampling, isotopologue analysis, and pool-size measurements, while true flux inference requires explicit modeling and precursor-enrichment assumptions. Spatial metabolomics extends this logic into tissue context by combining metabolite imaging, histology, cellular phenotyping, and isotope tracing to resolve which cells, where, and from which precursor metabolic activity arises.

4.7 New frontiers: single-cell, population-scale, and spatially integrated metabolomics (2026)

A 2026 Nature perspective describes two complementary scales of metabolomics: single-cell measurements that reveal cellular heterogeneity and population-scale profiling that supports studies of disease and environmental exposures [R16]. A separate 2026 Nature Methods study introduced SpaMTP, an analysis framework within a Seurat-oriented architecture for integrating spatial metabolomics and transcriptomics [R17]. The pedagogical link is direct: cell-resolved metabolic features and transcript localization can motivate enzyme/receptor hypotheses, but integration cannot correct a wrong compound annotation, mismatched tissue coordinates or missing experimental controls.

5 | Data analysis, statistics, and multi-omics integration

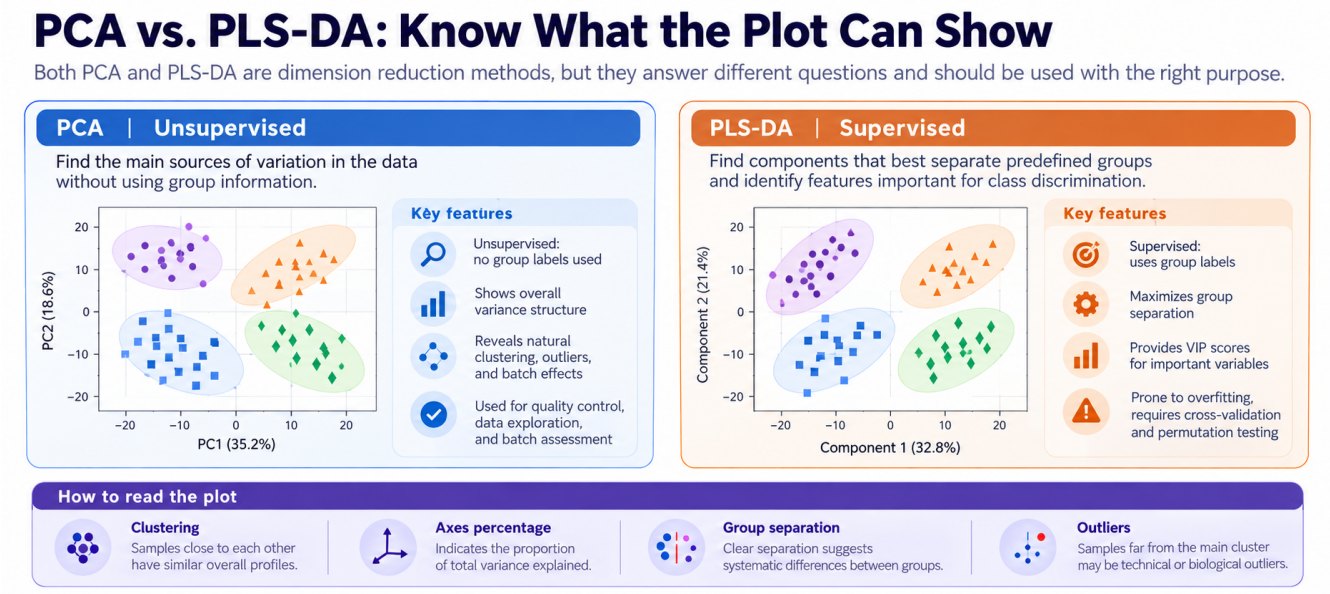


Figure 17. PCA and PLS-DA answer different questions. Exploratory visualization does not replace validation, and supervised separation requires explicit cross-validation and permutation controls.

For both proteomics and metabolomics, statistical analysis has three layers. The first layer is descriptive and exploratory: data distributions, missingness, heatmaps, correlation structure, PCA, sample clustering, and batch inspection. The second layer is inferential: differential abundance or concentration testing, covariate adjustment, effect sizes, false-discovery-rate correction, and confidence intervals. The third layer is biological interpretation: pathway enrichment, network analysis, prior knowledge integration, and the formulation of a falsifiable next experiment.

Students should be able to explain why PCA and PLS-DA are not interchangeable. PCA is unsupervised and useful for outlier detection, QC evaluation, and broad variance structure. PLS-DA is supervised and may show attractive group separation even when generalization is weak; therefore, it requires cross-validation, permutation testing, and cautious interpretation. Similarly, a volcano plot summarizes a statistical comparison but does not itself prove mechanism. A heatmap visualizes selected patterns but does not validate them. Every plot must be linked to a biological question and a statistical design.

5.1 Statistics: what exploratory and supervised models can demonstrate

Before differential analysis, make plots of blanks, pooled QCs, internal-standard response, total detected features, missingness and retention-time drift across run order. For untargeted features, exclude unstable signals according to a predeclared QC rationale and distinguish analytical correction from biological normalization. Probabilistic quotient normalization, median normalization and internal-standard scaling address different assumptions; a workflow should document why a particular method is suitable. “KNN for all missing data” and “one normalized signal per compound” are not universal rules.

PCA projects variation without using group labels; group separation on a PCA plot is visual evidence of structured variation, not a test of biological group difference. PLS-DA uses labels and can overfit; do not use a VIP threshold alone as biomarker validation. Use train/test splits or appropriately nested cross-validation, fully encapsulate preprocessing and feature selection inside training folds, and report performance on an independent set whenever possible. For each metabolite, predefine the comparison, estimate effect size with confidence interval, adjust for multiple testing, and control clinical or batch covariates. Pathway enrichment must use chemically supported identities and an appropriate measured-metabolite universe; unidentified m/z features should not be force-assigned to whichever pathway appears most appealing.

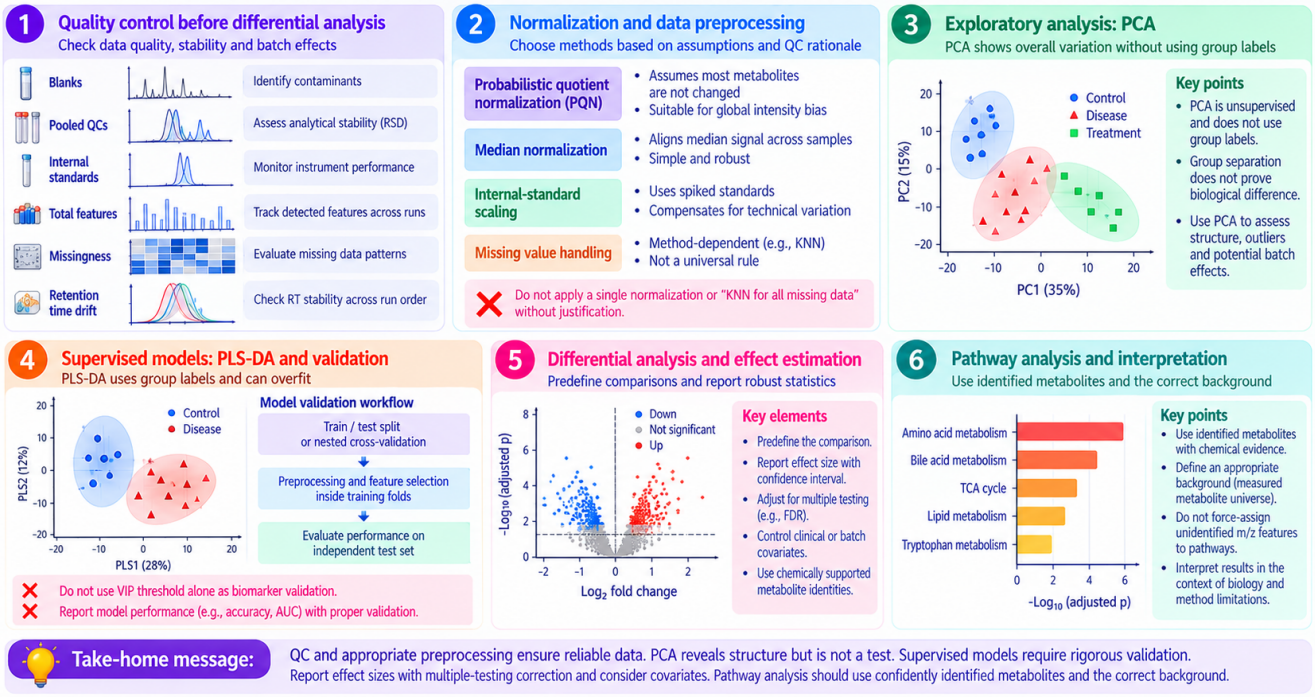


Figure 18. Statistical analysis of proteomics and metabolomics data: from QC to validated inference. Exploratory analysis should first evaluate blanks, pooled QCs, internal standards, missingness, run-order drift, and the assumptions behind normalization before testing biological differences. PCA visualizes unlabeled variation, whereas supervised models such as PLS-DA require leakage-free cross-validation and independent validation; differential testing should report effect sizes, confidence intervals, multiple-testing correction, covariate adjustment, and pathway analysis only for chemically supported metabolite identities.

5.2 Multi-omics integration: linking measurements with mechanisms

The most important conceptual upgrade for students is to stop viewing multi-omics integration as "put all matrices together and let software find the answer." A stronger way to think is to start from the scientific question, define which layer is expected to carry upstream information, which layer is expected to carry function, and which layer is closest to the phenotype. Then choose the integration method that matches that logic.

5.2.1 Matched samples, compartment, scales, and study design

The first task is not modeling; it is a sample map. Distinguish paired multi-omics (the same biological unit sampled for multiple assays) from unpaired omics (different cohorts, time points or biopsies). Preserve animal, patient and culture identifiers, technical replicate IDs, tissue compartment, fasting state, collection time and batch. The microbiome may be compositional, RNA-seq starts as counts, MS-based proteins are often log2 intensities, and targeted metabolite panels may have molar concentrations. These modalities cannot be concatenated as if they were identically scaled observations. Report which samples were excluded, which features were collapsed to genes or pathways, and whether repeated measures were represented in the statistical design.

5.2.2 Pairwise, pathway-based, factor-based, and supervised integration

Pairwise cross-omics correlations provide a transparent first view, but a microbiome-metabolite correlation can be confounded by diet, age, sequencing depth, microbial compositional structure or the host organ. For metagenomic relative-abundance data, use compositionality-aware transformations or models rather than naively applying Pearson correlation to sparse percentages. For repeated measures, account for subject-level dependence. After filtering to biologically interpretable features, test an explicit limited family of associations and apply multiple-testing correction.

Pathway-centered integration maps transcripts, proteins and metabolites to reactions using organism-aware identifiers. Discordance is biologically informative: a transcript and enzyme may rise while a product falls if substrate delivery or downstream consumption changes. Latent-factor approaches such as MOFA2 can identify shared variation across multi-omics blocks without requiring a phenotype label, but a latent factor is not itself a pathway and must be interpreted through loadings and measured covariates. Supervised multi-block approaches such as DIABLO can prioritize discriminating signatures; training and cross-validation must account for every data block, and feature stability should be assessed. Network diagrams are hypotheses about relations, not proof of physical binding or metabolic conversion.

5.2.3 Association, candidate mediator, causal mechanism, and intervention

An observational association supports co-variation. A replicated temporal sequence supports ordering but can retain confounding. Stable-isotope tracing may constrain the origin of a metabolite, and a loss-of-function experiment can test necessity under specific conditions. Dose-response and add-back rescue can test sufficiency or partial mediation, while target deletion or receptor antagonism tests pathway dependence. A full mechanistic story needs controls that distinguish direct host effects, off-target treatment effects, microbial community changes and compositional shifts. A mediation model fitted to cross-sectional data is not automatically proof that the proposed mediator is causal.

Students should formulate a hypothesis in the form “If upstream factor X changes the production of confirmed mediator M, then host protein response P and phenotype Y should shift; preventing M formation or blocking receptor R should attenuate the effect, and controlled restoration of M should recover it.” This formulation identifies testable interventions and potential falsifying observations rather than simply assembling enriched pathways.

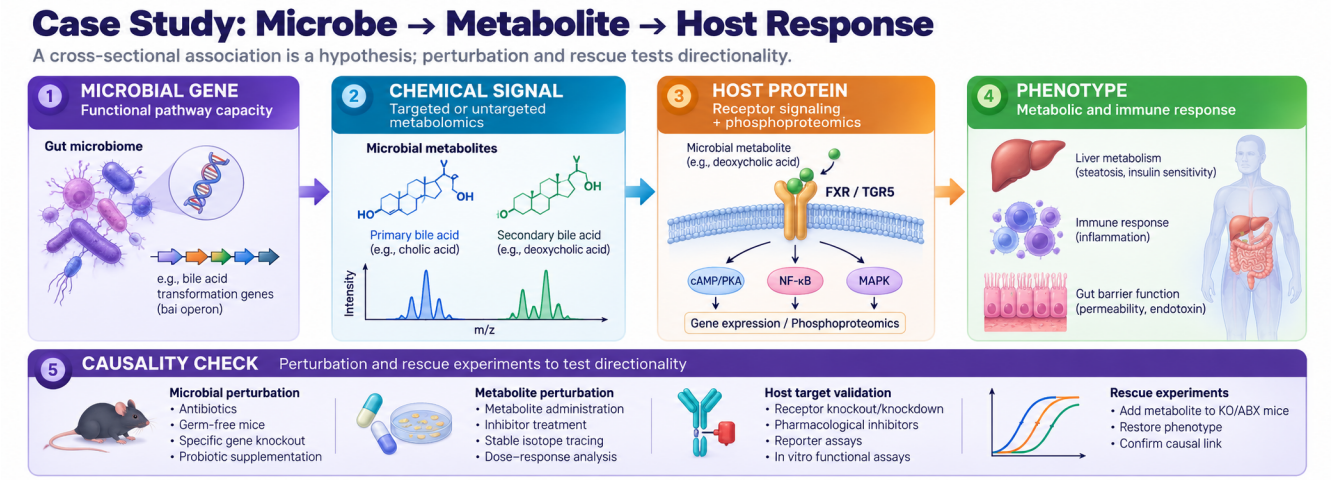


Figure 19. Example integration logic for a host-microbiome hypothesis. Cross-sectional association is the beginning of the story; perturbation and rescue experiments are needed to test directionality and causality.

A useful teaching model is the microbe -> metabolite -> host protein -> phenotype chain. Suppose shotgun microbiome data indicate enrichment of secondary-bile-acid genes, targeted metabolomics shows increased deoxycholate, and host proteomics or phosphoproteomics indicates altered receptor signaling. This is already more informative than any single data type alone. Yet it is still an integrated hypothesis, not proof. The next step is to manipulate the microbial pathway, block the receptor, or add back the metabolite to test whether the predicted protein response and phenotype move in the expected direction.

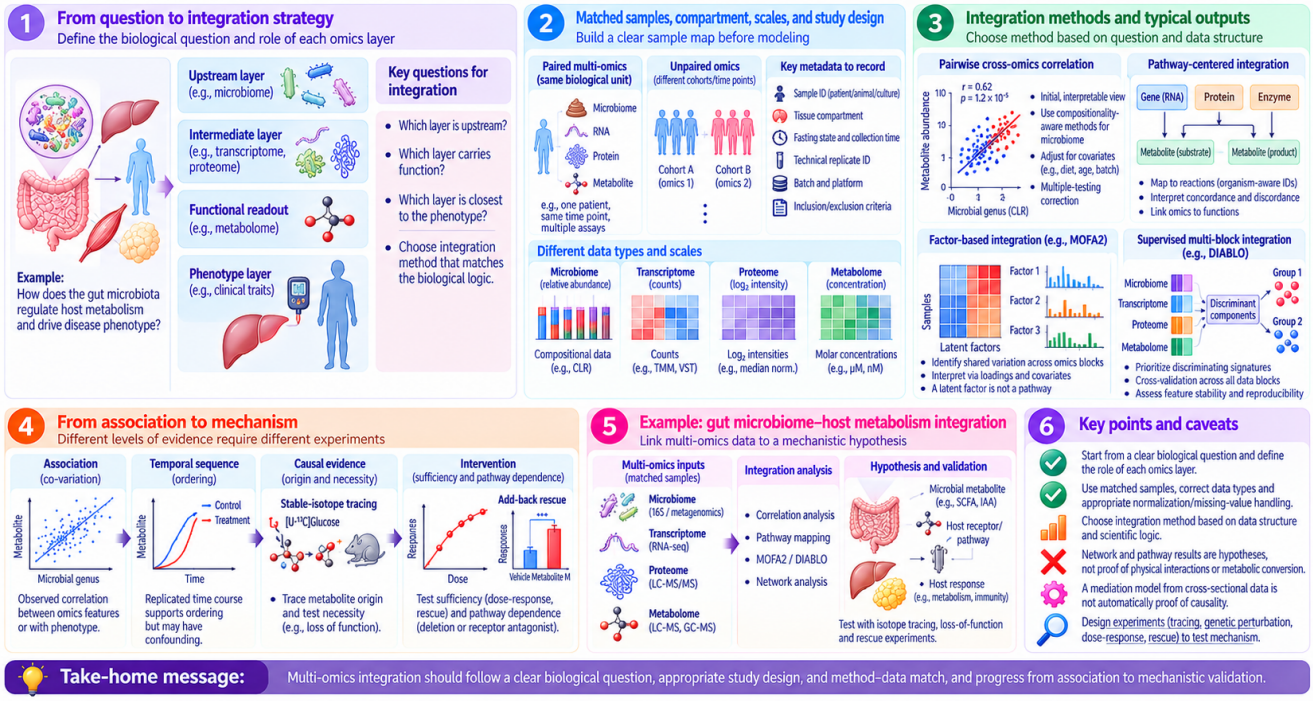


Figure 20. Multi-omics integration: linking measurements to mechanisms. Robust integration begins with matched sample design, modality-aware preprocessing, and explicit biological questions rather than simple matrix concatenation. Pairwise, pathway-based, latent-factor, and supervised approaches can

generate mechanistic hypotheses, but causal interpretation requires temporal evidence, tracing, perturbation, receptor/target validation, and rescue experiments that test the proposed X → M → P → Y pathway.

5.3 Frontiers in unpaired and spatially integrated omics (2026)

A 2026 Nature Biotechnology correspondence highlights the difficulties of integrating single-cell and spatial proteomics across unpaired datasets and platforms [R18]. For students, the practical decision is whether measurements derive from the same cell, adjacent registered tissue sections, the same specimen at different times, or entirely different cohorts. Each design supports a different level of inference. In particular, a strong integrated visualization is not proof that protein and metabolite signals coexist in the same cell unless the assay and registration strategy establish that correspondence.

6 | Translational research applications: synthetic biology, biopharma, and health

6.1 Synthetic biology: Design-Build-Test-Learn of a microbial metabolite

Imagine engineering a bacterial strain to increase the production of an anti-inflammatory indole derivative. Design: use genome annotation and pathway reconstruction to identify enzymes, competing routes, transporters and cofactors; construct a genome-scale metabolic model as a hypothesis about stoichiometric capacity, not a guarantee of actual flux. Build: edit the promoter, coding sequence or pathway architecture in a defined host and maintain sequence-confirmed strain identity. Test: use targeted LC-MS/MS with authentic standards to quantify the desired product and competing by-products over a calibrated fermentation time course. Run proteomics to determine whether the engineered enzymes are expressed at appropriate abundance and whether stress-response or secretion pathways become bottlenecks. Use isotope tracing or uptake/secretion rates if an origin or flux claim is required.

Learn: integrate RNA-protein discordance, enzyme levels, growth rate, substrate use, product titer, yield and productivity. For example, unchanged product titer despite higher enzyme abundance could indicate insufficient substrate, transport limitation, cofactor imbalance, enzyme inactivation or product degradation. A next DBTL iteration should change one or a small factorial set of mechanistically justified variables, document replicate variation and compare strains on both biological performance and burden to the host. AI can help propose design candidates or rank experiments, but model predictions must be constrained by actual assay data and verified molecular sequences.

6.2 Biopharmaceuticals: targets, mechanism, and precision medicine

A clinical biomarker pipeline begins with a patient-relevant phenotype and a well-defined sampling process. Broad proteomics can discover candidate proteins, but disease association does not establish a diagnostic test. Select a parsimonious panel using leakage-free modeling, follow up with targeted proteomics or orthogonal immunoassays, and assess calibration, discriminatory performance, confidence intervals, clinical covariates and performance in an independent population. Circulating-protein origins may be inferred only cautiously; organ-specific expression is not the same as exclusive tissue origin. In a 2024 Nature Medicine analysis involving 41,931 participants, broad plasma proteomics was integrated with clinical information to study prospective risk signatures across multiple diseases [R10]. It illustrates the need to distinguish promising prediction signals from established clinical utility.

Proteogenomics links tumor DNA/RNA variation to antigen presentation and can prioritize immunotherapy targets; immunopeptidomic evidence does not itself establish T-cell recognition. Spatial proteomics plus metabolomics can stratify tumor regions, but this requires cell segmentation, image registration and careful spatial inference. Drug target engagement studies may combine affinity capture/thermal profiling, phosphoproteomics and metabolomics to distinguish proximal molecular changes from downstream metabolic consequences. Therapeutic claims require independent target perturbation and clinically relevant endpoints.

6.3 Life-course health: microbiome-gut-liver-muscle-adipose communication

In metabolic health and aging, a plausible study might combine stool metagenomics, fecal and plasma bile-acid panels, muscle/adipose phosphoproteomics and serum cytokine/secretome proteomics, with phenotypes including insulin sensitivity, tissue morphology and mitochondrial function. A cross-tissue signal should be traced through source, circulation, receptor and effect. Exogenous administration of a microbiome-derived metabolite is not equivalent to demonstrating that the endogenous microbiome is its primary source in vivo; isotope tracing, microbial pathway perturbation and host genetic models can disentangle these possibilities. Aging-related plasma protein signatures can provide useful descriptive or predictive associations, but biological age estimates are model outputs rather than direct measures of the molecular mechanism of aging. A 2025 Nature Medicine study analyzed 2,916 plasma proteins in 44,498 UK Biobank participants to estimate organ-specific age measures and examine their associations with health outcomes [R11]. For undergraduate research design, the next question is how tissue-specific proteomics, spatial cell-state mapping and targeted metabolites could test a mechanistic mediator suggested by an association.

6.4 Environmental, agricultural, and microbial biotechnology

The same integration logic applies beyond human stool: soil, freshwater, wastewater and fermentation consortia can be profiled by marker sequencing or metagenomics for membership/potential; metaproteomics tests expressed enzymes; exometabolomics characterizes compounds that accumulate or disappear; and chemical assays measure the environmental endpoint of interest. Stable-isotope probing can link a substrate to active organisms or metabolite pools, but taxonomic assignment of a shared peptide and origin inference remain difficult in diverse environmental communities. Synthetic consortia require defined input ratios, culture conditions, substrate mass balance and repeated stability measurements rather than relying on an attractive co-occurrence graph.

7 | Reading primary papers: from method to hypothesis

Paper-reading exercise: for every article above, write four sentences: (1) the biological gap, (2) the measurement innovation, (3) the strongest reported observation with the real population/model, and (4) a plausible next experiment that would distinguish an association from a causal mechanism. Do not copy a methods headline into a stronger biological claim: "identified proteins in single HeLa cells" cannot be generalized automatically to every clinical biopsy, and "spatial metabolite signal" is not automatically a uniquely confirmed structure.

Primary study	Technology advancement and sample context	What it supports / unresolved question
Chip-Tip single-cell proteomics (2025) [R2]	Low-loss single-cell preparation + high-sensitivity nDIA	Deep protein identification in a defined HeLa system; assess transfer to scarce tissue cells
Spatial metabolite + immune imaging (2024) [R8]	Same-section MALDI-MSI + imaging mass cytometry	Cell-type-associated metabolite maps; pixel mixing and chemical ambiguity remain
irCLIP-RNP / Re-CLIP (2025) [R5]	CLIP coupled to MS / sequential RBP capture	Dynamics of RNA-centered protein assemblies; binding alone is not phenotype
NeoDisc proteogenomics (2025) [R6]	WGS/RNA + HLA immunopeptidomics	Presented peptide prioritization; immune recognition requires functional testing
Plasma proteomic risk models (2024) [R10]	Large prospective cohort + broad plasma panels	Association/prediction; external transportability and utility remain separate
Spatial 13C tracing (2024) [R9]	MALDI-AIF + isotope modeling	Cellular heterogeneity in precursor contribution; absolute flux needs more constraints

MZmine 3 protocol (2024) [R7]	Reproducible LC-IMS-MS/GC/MSI feature extraction	Accessible processing workflows; parameter and annotation QC still required

8 | AI agents in functional omics: reproducible operations and human judgment

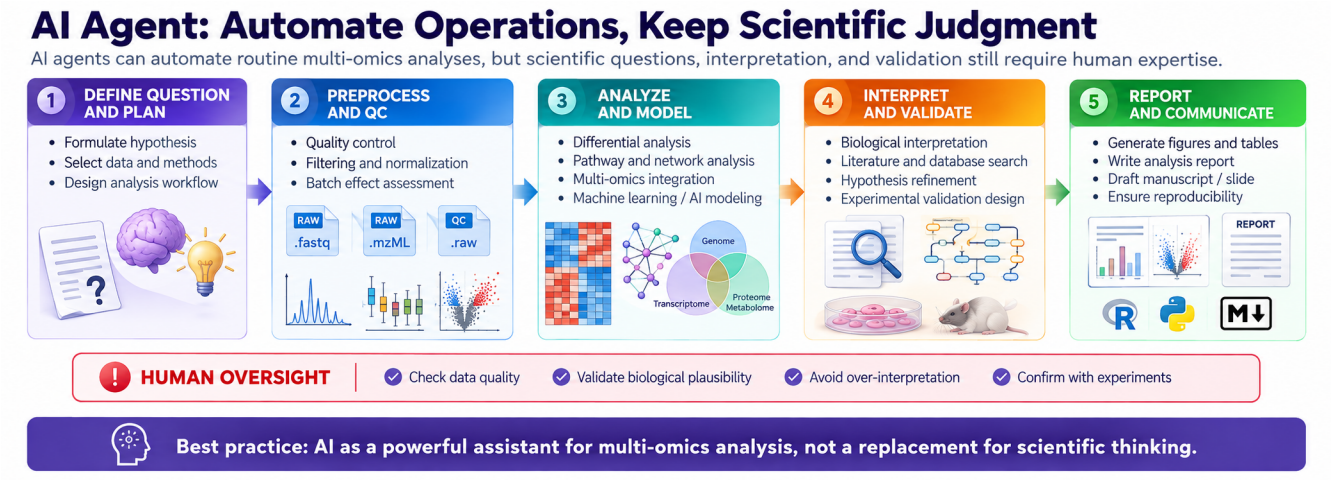


Figure 21. AI-agent workflow. Automation is useful only when the run remains auditable, reproducible, and subject to explicit human checkpoints.

The goal of AI in this course is not to let students "throw the data to a chatbot." The real goal is to teach a disciplined workflow in which the student defines the question, the AI agent helps operationalize the analysis, and the student then audits every important assumption and claim. In this setting, an agent is not just a language model. It is a tool-connected system that can read files, generate code, call databases, produce plots, keep logs, and revise its own steps under supervision.

Different agent ecosystems have different strengths. ChatGPT-based workflows are often effective for structured reasoning, code generation, and interactive analysis conversations. Gemini-based workflows are strong for long-context reading of PDFs, multimodal interpretation, and integration with Google-linked environments. Claude-based workflows are often strong for long-context drafting, careful instruction following, and coding or document-centered analysis. In teaching, the correct question is not "Which one is best?" but "Which tasks can be safely delegated, and how will I verify the output?"

The most robust prompts are explicit. Ask the agent to show its assumptions, keep a run log, separate code from interpretation, and flag every part of the workflow that depends on unverified identifiers or incomplete annotation. This protects students from one of the most common AI-related errors: a fluent but overconfident analysis narrative that skips critical QC or misstates evidence strength.

8.1 How ChatGPT-, Gemini-, and Claude-based workflows can assist

An AI agent is a language model connected to tools and files under a defined permission model. A plain chat response may draft code but does not prove execution; a reproducible workflow records the exact input files, analysis parameters, program versions, commands, execution logs, QC figures and all resulting tables. ChatGPT-based data analysis can work with uploaded tables and run Python computations in supported configurations; Gemini's documented API code-execution tool runs Python, not native R execution; Claude-based workflows may write R scripts, inspect files and use permitted execution tools depending on the product and environment. For the course, all three can help author R scripts, while actual R execution must occur in an R-enabled local or connected environment; do not assume that a platform has a licensed scientific database or a configured R executor simply because it can describe one.

The most reliable division of labor is: student defines hypotheses and biological units; AI drafts a plan and checks input schemas; R/Bioconductor or specialized mass-spectrometry software processes real data; AI helps compare log files and literature; student audits normalization, uncertainty and annotation; wet-lab experiments validate prioritized claims. Permissions should be limited to project directories and public/reference data where feasible. Human clinical identifiers should not be sent to external services without appropriate approvals and de-identification.

8.2 Six human checkpoints for agent-executed analyses

- Gate 0 — research question: specify exposure, comparison, outcome, biological replicate, tissue, time and plausible confounders before the agent opens any matrix.
- Gate 1 — input integrity: check hashes, sample IDs, expected dimensions, controls, nonnegative concentrations, units and run metadata; stop on unmatched or duplicated IDs.
- Gate 2 — assay-aware QC: require different metrics for DIA/PRM, Co-IP, untargeted LC-MS, targeted calibration and metagenomic relative abundance.
- Gate 3 — statistics: define test family, normalization, covariates, multiple testing and out-of-fold preprocessing before model training.
- Gate 4 — evidence: separate measured proteins, measured metabolite features, confidently assigned chemicals and database-predicted interactions.
- Gate 5 — output: deposit executed scripts, session information, tables, figures, a limitations paragraph and a proposed independent experiment.

8.3 Domain-specific AI-agent prompts

Prompt template for proteomics agent:

Context: You are assisting with a university-level proteomics analysis. I have a protein-by-sample matrix exported from MaxQuant, FragPipe, or DIA-NN and a metadata table.
Action: First inspect the metadata and report the biological contrast, covariates, and possible batch variables. Then write reproducible R code to (1) filter contaminants and low-quality proteins, (2) summarize missingness and sample-level QC, (3) normalize or justify why no additional normalization is needed, (4) fit the correct differential-abundance model, (5) control BH-FDR, (6)

generate PCA, volcano, and heatmap outputs, and (7) return a table of significant proteins with effect sizes and annotations.
Result: Produce code, plots, and a short report. Keep QC interpretation separate from biological interpretation.
Validation: Do not claim mechanism from a volcano plot alone. Flag ambiguous identifiers, batch effects, or high missingness before continuing.

Prompt template for metabolomics agent:

Context: You are assisting with an untargeted or targeted metabolomics analysis. I have a feature table or targeted concentration matrix and sample metadata.
Action: (1) Check blanks, pooled QCs, internal standards, run order, and missingness; (2) filter low-quality features; (3) choose an appropriate normalization and transformation strategy; (4) run PCA and, if requested, a validated supervised model; (5) perform differential testing with multiple-testing correction; (6) separate annotation confidence from biological interpretation; (7) summarize pathways only after stating the evidence level of identified metabolites.
Result: Provide code, plots, and a ranked result table.
Validation: Never present a mass-only match as a confirmed metabolite. Explicitly report whether a result is feature-level, putative annotation, or authentic-standard confirmed.

Prompt template for multi-omics integration agent:

Context: I have matched proteomics, metabolomics, and optional microbiome or transcriptomics data from the same samples.
Action: (1) Harmonize sample IDs; (2) summarize missingness per block; (3) propose pairwise and multi-block integration analyses; (4) execute a transparent integration pipeline using correlation analysis plus either mixOmics/DIABLO or MOFA2; (5) identify coordinated features linked to the phenotype; (6) generate an integration network and pathway summary.
Result: Produce code, plots, and a concise explanation of what is associative versus what is experimentally testable.
Validation: Distinguish correlation, prediction, and causality. Do not infer directionality unless supported by design or perturbation.

8.4 A reusable CARE + validation prompt for multi-omics projects

Reusable student-controlled prompt: paste into ChatGPT, Gemini or Claude and attach files only after removing identifiers not needed for the analysis.

8.5 Boundaries of specialized agents and source-grounded claims

For proteomics, the agent may organize FragPipe, MaxQuant or DIA-NN exports, but it must not claim that changing a summary table is equivalent to reprocessing raw spectra. For metabolomics, an agent may construct SIRIUS candidate lists or parse GNPS/MZmine exports, but it must not assign a unique structure from accurate mass alone. A text-based search of HMDB should not be described as a formal authenticated API call unless the endpoint and response are actually verified. For multi-omics, every inferred feature correspondence should be saved in an ID mapping file, and network arrows should reflect documented reaction topology or an explicit experimental design rather than the sign of correlation.

```
CONTEXT: I study [phenotype] in [species / tissue / cohort]. The supplied
files are [metadata.csv] and [assay matrix]. State what the assay actually
measures; do not guess missing identifiers or sample sizes.
ACTION: Inspect the schema and propose QC, transformations, statistics and
multi-omics integration. Stop and ask me to approve exclusions, covariates,
missing-value strategy and the exact contrast BEFORE modeling.
RESULT: Generate an R script, expected output manifest, and source-backed
method notes. Return uncertainty, effect sizes, FDR and annotation levels.
EXAMPLE: A microbiome gene → chemically confirmed bile acid → host
phosphosite → phenotype is a candidate chain, not an established mechanism.
VALIDATION: Execute only with authorized tools; return logs and sessionInfo.
Never invent a DOI, API return value, p-value, metabolite identity, observed
protein interaction or successful execution. Mark unexecuted code clearly.
```

9 | Reproducible R practice: validated inputs and assay-aware analysis

The following R scripts illustrate matrix-level analysis with explicit sample and input validation. Each block specifies an input schema and validation checks; exported matrix-level analysis does not substitute for reprocessing raw spectra. Adapt the design, covariates, and missingness approach to the experiment.

9.1 Project directory, sample manifest, and data integrity

I.1: real-data input contract; never match samples by assumed position or name-sorting alone.

```
# Run in an R-enabled environment. Input: metadata.csv,
# protein_matrix.csv, metabolite_matrix.csv.
# Matrices: first column FeatureID; remaining columns EXACT SampleID.
# Metadata: SampleID, Group, Batch; optional Age, Sex, BMI.
options(stringsAsFactors = FALSE)
required <- c('metadata.csv', 'protein_matrix.csv', 'metabolite_matrix.csv')
stopifnot(all(file.exists(required)))
meta <- read.csv('metadata.csv', check.names = FALSE)
stopifnot(all(c('SampleID','Group','Batch') %in% names(meta)))
```

```
stopifnot(!anyNA(meta$SampleID), !anyDuplicated(meta$SampleID))
stopifnot(all(nzchar(meta$SampleID)), !anyNA(meta$Group))
read_block <- function(path, meta) {
  x <- read.csv(path, check.names = FALSE)
  stopifnot('FeatureID' %in% names(x), !anyDuplicated(x$FeatureID))
  stopifnot(all(meta$SampleID %in% names(x)))
  x <- x[, c('FeatureID', meta$SampleID), drop = FALSE]
  z <- as.matrix(x[,-1,drop=FALSE]); storage.mode(z) <- 'numeric'
  rownames(z) <- x$FeatureID
  stopifnot(identical(colnames(z), meta$SampleID))
  z
}
prot <- read_block('protein_matrix.csv', meta)
met  <- read_block('metabolite_matrix.csv', meta)
print(list(n_samples=nrow(meta), n_proteins=nrow(prot),
           n_metabolite_features=nrow(met)))
write.csv(meta, 'validated_metadata.csv', row.names = FALSE)
```

9.2 Proteomics: QC, missingness-aware limma, audit tables

I.2: illustrates abundance testing with incomplete observations; inspect software-specific quality flags before using the exported matrix.

```
suppressPackageStartupMessages({library(limma); library(ggplot2)})
stopifnot(all(is.na(prot) | prot >= 0))
prot[prot == 0] <- NA  # ONLY if instrument export documents zero = not detected
prot <- prot[rowSums(is.finite(prot)) >= 3, , drop=FALSE]
qc <- data.frame(SampleID=colnames(prot),
                 Detected=colSums(is.finite(prot)),
                 MissingFraction=colMeans(!is.finite(prot)))
write.csv(qc, 'protein_sample_qc.csv', row.names=FALSE)
# For abundance analysis, log-transform raw positive intensities.
# If vendor export is already log2-normalized, skip this transformation.
logp <- log2(prot)
# limma tolerates missing observations; do not impute by default.
# Fit only proteins with enough observed samples in BOTH groups.
meta$Group <- factor(meta$Group, levels=c('Control','Treatment'))
stopifnot(!anyNA(meta$Group), length(unique(meta$Group))==2)
meta$Batch <- factor(meta$Batch)
obs <- vapply(split(seq_len(nrow(meta)),meta$Group), function(ii)
  rowSums(is.finite(logp[,ii,drop=FALSE])), numeric(nrow(logp)))
keep <- rowSums(obs >= 2) == 2
logp <- logp[keep,,drop=FALSE]
```

```
stopifnot(nrow(logp)>1)
# Median centering is OPTIONAL and assumes broad global changes are limited.
med <- apply(logp, 2, median, na.rm=TRUE)
logp <- sweep(logp, 2, med - median(med), '-')
# Avoid batch term if it is exactly confounded with Group.
design <- model.matrix(~ Group + Batch, meta)
stopifnot(qr(design)$rank==ncol(design))
fit <- eBayes(lmFit(logp, design))
coef_name <- grep('^GroupTreatment$', colnames(design), value=TRUE)
stopifnot(length(coef_name)==1)
res <- topTable(fit, coef=coef_name, number=Inf, adjust.method='BH')
res$FeatureID <- rownames(res)
write.csv(res, 'protein_differential.csv', row.names=FALSE)
res$Class <- ifelse(res$adj.P.Val < .05 & abs(res$logFC)>1,
                    ifelse(res$logFC>0,'Up','Down'),'Other')
p <- ggplot(res,aes(logFC,-log10(pmax(adj.P.Val,1e-300)),colour=Class))+
  geom_point(alpha=.65,size=1)+theme_bw()+
  labs(x='log2 fold change',y='-log10 BH-adjusted p')
ggsave('protein_volcano.pdf',p,width=7,height=5)
```

9.3 Targeted metabolites: calibration units, paired data, multiple tests

I.3: quantitative differences in the unit provided by calibration; do not relabel an uncalibrated feature intensity as micromolar.

```
# Expected: targeted_concentrations.csv rows=metabolites,
# SampleID columns; values already calibrated in one documented unit.
tgt <- read_block('targeted_concentrations.csv', meta)
stopifnot(all(is.na(tgt) | tgt>=0))
# Group contrast, adjust for Batch and optional numeric age, if present.
meta$Group <- factor(meta$Group,levels=c('Control','Treatment'))
meta$Batch <- factor(meta$Batch)
covars <- c('Group','Batch', if('Age' %in% names(meta)) 'Age')
form <- as.formula(paste('y ~',paste(covars,collapse=' + ')))
ans <- lapply(seq_len(nrow(tgt)),function(i) {
  df <- meta;df$y <- as.numeric(tgt[i,])
  df <- df[is.finite(df$y),,drop=FALSE]
  if(nrow(df)<6 || length(unique(df$Group))<2) return(NULL)
  m <- try(lm(form,df),silent=TRUE)
  if(inherits(m,'try-error')) return(NULL)
  cc <- summary(m)$coefficients
  if(!('GroupTreatment' %in% rownames(cc))) return(NULL)
  data.frame(FeatureID=rownames(tgt)[i],
```

```
        Difference=unname(cc['GroupTreatment','Estimate']),
        P=unname(cc['GroupTreatment','Pr(>|t|)']))
})
ans <- do.call(rbind,ans)
stopifnot(!is.null(ans),nrow(ans)>0)
ans$FDR <- p.adjust(ans$P,method='BH')
write.csv(ans,'targeted_metabolite_differences.csv',row.names=FALSE)
# Difference is in ORIGINAL concentration units, not log2FC.
```

9.4 Protein-metabolite associations using matched samples

I.4: replace illustrative candidate names by exact validated IDs; adapt to subject, batch and compositional covariates for a real study.

```
# Avoid all-by-all correlation fishing. Use prespecified candidate sets.
candidate_proteins <- c('Protein_A','Protein_B')
candidate_metabolites <- c('Metabolite_1','Metabolite_2')
P0 <- logp[intersect(candidate_proteins,rownames(logp)),,drop=FALSE]
M0 <- tgt[intersect(candidate_metabolites,rownames(tgt)),,drop=FALSE]
stopifnot(nrow(P0)>0,nrow(M0)>0)
stopifnot(identical(colnames(P0),colnames(M0)))
output <- list()
for(i in seq_len(nrow(P0))) for(j in seq_len(nrow(M0))){
 a <- as.numeric(P0[i,]);b <- as.numeric(M0[j,])
 ix <- is.finite(a)&is.finite(b)
 if(sum(ix)<8) next
 ct <- suppressWarnings(cor.test(a[ix],b[ix],method='spearman',exact=FALSE))
 output[[length(output)+1]] <- data.frame(
  Protein=rownames(P0)[i],Metabolite=rownames(M0)[j],
  N=sum(ix),Rho=unname(ct$estimate),P=ct$p.value)
}
output <- do.call(rbind,output)
if(!is.null(output)){
 output$FDR <- p.adjust(output$P,method='BH')
 write.csv(output,'prespecified_cross_omics_links.csv',row.names=FALSE)
}
# Correlation direction never alone establishes a causal arrow.
```

9.5 Unsupervised factors and carefully validated multiblock learning

I.5: intentionally stops before model fitting to enforce supervised validation rather than present untested tuning defaults as a complete analysis.

```
# Optional advanced module; requires installed Bioconductor / CRAN packages.
# Matrices are features x samples; scale blocks only AFTER sample matching.
# MOFA2 syntax may vary with installed release; inspect package docs.
if(requireNamespace('MOFA2',quietly=TRUE)) {
  stopifnot(identical(colnames(logp),colnames(tgt)))
  blocks <- list(Proteome=logp, Metabolome=tgt)
  # Real work: filter low-variance features and transform metabolite
  # concentrations using an assay-specific policy BEFORE this call.
  message('MOFA2 input blocks ready; create_mofa(blocks) after QC approval.')
}
if(requireNamespace('mixOmics',quietly=TRUE)) {
  # DIABLO needs sample x feature matrices and an outcome vector.
  X <- list(Proteome=t(logp),Metabolome=t(tgt))
  y <- factor(meta$Group)
  stopifnot(all(vapply(X,nrow,1L)==length(y)))
  # Hyperparameter selection and all preprocessing must occur within
  # the training folds; do not report training separation as performance.
  message('DIABLO input prepared; define nested CV and tune keepX first.')
}
```

10 | Scientific reasoning: what each result supports

1. A measured protein difference is not automatically a functional mechanism; ask whether the protein's activity, localization, complex membership, or PTM state is also relevant.
2. A detected metabolite feature is not automatically a confirmed compound; ask what evidence supports the name.
3. A correlation between a microbe and a metabolite is not automatically causality; ask whether the pathway, chemistry, and perturbation evidence agree.
4. A predictive multi-omics model is not automatically biologically interpretable; ask whether the selected features map onto coherent mechanisms.
5. An AI-generated analysis is not automatically trustworthy; ask whether every key step is documented, reproducible, and supported by real source evidence.

11 | Integrated practice questions

1. List five quality-control checks you would inspect in a proteomics matrix before drawing a volcano plot, and five analogous checks for metabolomics.
2. Why is PCA useful even though it is not a significance test? Why is PLS-DA potentially dangerous if used without validation?

3. Write a short CARE + Validation prompt to instruct an AI agent to analyze a targeted bile-acid dataset while preserving annotation confidence and statistical rigor.
4. When should you choose PRM/SRM/MRM rather than global DIA proteomics?
5. Why might multi-omics integration reveal a stronger hypothesis than any single omics layer alone, but still fall short of causality?
6. A regulatory variant increases an mRNA but not its protein. Specify one assay each for RNA structure, protein turnover and PTM activity. Give a falsifying observation for your preferred explanation.
7. Design an untargeted-to-targeted pipeline for a host-microbe bile-acid mediator. Specify sample collection, reference standards, MS2/retention evidence, and a rescue experiment.
8. Co-IP-MS identifies a candidate partner and TurboID detects it in a membrane neighborhood. Explain which conclusions are justified and describe two controls for each assay.
9. In a time-course phosphoproteomics study, the site signal doubles but the parent protein also doubles. How would you evaluate whether site occupancy or pathway activity changed?
10. A stool metagenome suggests butyrate-production capacity but serum butyrate is unchanged. Propose explanations involving substrates, local production/consumption, transport and sampling.
11. In a fermenter, product titer rises but yield on glucose falls. Explain why metabolomics, proteomics and carbon balance can suggest different bottlenecks.
12. Design a comparison of label-free DIA, TMT and a PRM panel for a 200-sample plasma cohort. State sample randomization, reference strategy, cost/coverage tradeoffs and validation plan.
13. An unknown LC-MS feature has an accurate mass hit for three isomers and no reference MS2. Write a defensible Results sentence and an experiment to raise identification confidence.
14. A PLS-DA plot separates disease and control but all disease samples were acquired in one MS batch. Explain the confounding and a redesign that avoids information leakage.
15. An AI agent reports a dramatic microbe-metabolite-protein pathway with one invalid HMDB ID. Write a provenance audit and decide which evidence can be retained.
16. Using a specified ncRNA, construct an experiment integrating RNA-seq, RIP/eCLIP, RNA pull-down-MS, total proteomics and functional rescue; specify what each assay alone cannot show.
17. Write a five-link causal hypothesis joining DNA regulation, RNA, an enzyme, a metabolite and phenotype. Label direct measurements, inferred relationships and interventions.

12 | References and reporting resources

Selected foundation and recent primary literature [R1-R18]. Study-specific sample sizes, coverage and findings are reported in their original contexts; see the linked papers for methods, cohort definitions and limitations.

[R1] Guo T, Steen JA, Mann M. Mass-spectrometry-based proteomics: from single cells to clinical applications. Nature 638, 901-911 (2025). DOI: https://doi.org/10.1038/s41586-025-08584-0

[R2] Ye Z et al. Enhanced sensitivity and scalability with a Chip-Tip workflow enables deep single-cell proteomics. Nature Methods 22, 499-509 (2025). DOI: https://doi.org/10.1038/s41592-024-02558-2

[R3] Method of the Year 2024: spatial proteomics. Nature Methods 21, 2195-2196 (2024). DOI: https://doi.org/10.1038/s41592-024-02565-3

[R4] APEX2 and TurboID define unique subcellular proteomes. Scientific Reports (2025). DOI: https://doi.org/10.1038/s41598-025-27545-1

[R5] Ducoli L et al. irCLIP-RNP and Re-CLIP reveal patterns of dynamic protein assemblies on RNA. Nature 641, 769-778 (2025). DOI: https://doi.org/10.1038/s41586-025-08787-5

[R6] Huber F et al. A comprehensive proteogenomic pipeline for neoantigen discovery to advance personalized cancer immunotherapy. Nature Biotechnology 43, 1360-1372 (2025). DOI: https://doi.org/10.1038/s41587-024-02420-y

[R7] Heuckeroth S et al. Reproducible mass spectrometry data processing and compound annotation in MZmine 3. Nature Protocols 19, 2597-2641 (2024). DOI: https://doi.org/10.1038/s41596-024-00996-y

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[R9] Buglakova E et al. Spatial single-cell isotope tracing reveals heterogeneity of de novo fatty acid synthesis in cancer. Nature Metabolism 6, 1695-1711 (2024). DOI: https://doi.org/10.1038/s42255-024-01118-4

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Software and reporting resources

MaxQuant — https://www.maxquant.org/

FragPipe — https://fragpipe.nesvilab.org/

DIA-NN — https://github.com/vdemichev/DiaNN

PRIDE / ProteomeXchange — https://www.ebi.ac.uk/pride/

XCMS — https://bioconductor.org/packages/xcms/

MZmine documentation — https://mzmine.github.io/mzmine_documentation/latest/

SIRIUS — https://bio.informatik.uni-jena.de/software/sirius/

MetaboAnalyst — https://www.metaboanalyst.ca/

Metabolomics Workbench — https://www.metabolomicsworkbench.org/

mixOmics — https://mixomics.org/

MOFA2 — https://biofam.github.io/MOFA2/

ChatGPT data analysis help — https://help.openai.com/en/articles/8437071-data-analysis-with-chatgpt

Gemini API code execution — https://ai.google.dev/gemini-api/docs/code-execution